

Implementing Tissue Heterogeneity in Finite Element Breast Models for Optical Cancer Detection

By

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Graduation report

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Summary

Breast cancer screening using X-ray mammography has reduced sensitivity in women with dense breast tissue, particularly those under 40 years of age, while concerns about cumulative radiation exposure highlight the need for radiation free alternatives. This study enhances the biomechanical realism of a finite element breast model developed for Early Warning Scan, a company working on a multi-camera optical imaging system designed to detect breast cancer by exploiting biomechanical differences between healthy and cancerous tissue.

The research implemented spatial heterogeneity in tissue material properties using a Python based computational pipeline integrated with FEBio 4.0 finite element solver and Gmsh mesh generator. A hemispherical breast geometry containing adipose and glandular tissue compartments was modelled with Mooney-Rivlin hyperelastic material properties. Heterogeneity was introduced through radial density and stiffness gradients in both tissue types, combined with discrete Gaussian lobular structures in the glandular region. Eight model variants were analysed during a simulated parabolic jump motion (1 cm vertical displacement), examining surface displacement patterns, internal stress distributions, and temporal dynamics through post processing of simulation output files.

The implementation of stiffness heterogeneity reduced peak vertical surface displacement by 13% compared to the homogeneous baseline (from 14.0 mm to 12.2 mm), while density gradients showed minimal effect (2% reduction). Landmark analysis revealed position dependent heterogeneity effects, with the nipple region exhibiting 89% displacement increase with stiffness gradients, whereas lateral positions showed consistent behaviour across models. Peak von Mises stress showed tissue specific responses: in glandular tissue, stiffness gradients reduced peak stress by 5% while density gradients increased it by 9%; in adipose tissue, both gradient types produced similar 17% stress reductions.

These findings demonstrate that spatial tissue property variations create detectable surface displacement differences that could enhance tumour detection in optical imaging systems. The 13% displacement change from a 28% stiffness gradient suggests that malignant tumours with their substantially higher stiffness contrasts (200-500% above surrounding tissue) should produce readily detectable displacement anomalies. The computational framework developed provides a foundation for generating synthetic training data for machine learning algorithms and supports future patient specific implementations incorporating medical imaging data.

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Introduction

Breast cancer remains the most common cancer among women worldwide, with early detection being crucial improving survival rates (WHO, 2022). The current gold standard screening method is mammography, which uses X-ray imaging. X-ray mammography has reduced sensitivity in women with dense breast tissue, making it less suitable for women under 40 years of age (von Euler-Chelpin et al., 2019). Additionally, concerns about cumulative radiation exposure, especially for younger women and those requiring frequent screening, highlighting the need for alternative, radiation free breast imaging technologies that can provide accurate detection across all breast tissue densities (RIVM, 2023).

A promising approach is multi camera imaging, developed by Early Warning Scan (EWS). This approach exploits the biomechanical differences between healthy tissue and cancerous tissue. Malignant tumors typically exhibit higher density and stiffness than surrounding breast tissue, which causes different deformation patterns during breast movement (Kim Joo Kim et al., 2025). By capturing these differences through synchronized multi camera imaging, this could potentially detect cancerous regions without radiation exposure. The development and validation of such imaging systems require realistic computation models that can simulate how breast tissue deforms under various conditions.

Finite element models (FEM) play a role in this development process. FEM allows researchers to simulate breast deformation and generate synthetic training data for artificial intelligence algorithms that will perform cancer detection (Sayed et al., 2020). For the AI to generalize effectively real clinical scenarios, the breast model must accurately represent the complex behavior of human breast tissue. Current FEM breast models often assume homogenous tissue values and simplified geometric structures, treating the entire breast as a uniform material with a perfect circular or spherical glandular region. This oversimplification fails to capture the true anatomical complexity of the breast.

Real breast tissue exhibits considerable heterogeneity in its mechanical properties. Adipose and glandular tissues differ in density and stiffness, with glandular tissue being approximately twice as stiff as adipose tissue (Teixeira & Martins, 2023). MRI elastography studies have demonstrated that fibroglandular tissue exhibits mean shear stiffness of 7.5 ± 3.6 kPa compared to 3.3 ± 1.9 kPa for adipose tissue, representing a double value difference (McKnight et al., 2002; Patel et al., 2020). Ex vivo mechanical testing of human breast tissue has shown standard errors of approximately 20% at peak deformation within tissue samples (Goodbrake et al., 2022). Furthermore, tissue properties vary spatially within each tissue type due to natural biological variability, gradual transitions at tissue boundaries, and regional density differences, with coefficients of variation ranging from 20-60% reported across elastography studies (Patel et al., 2020; Samani et al., 2007).

This project aims to enhance the biomechanical realism of an existing finite element breast model by implementing spatial heterogeneity in tissue material properties. Building on the EWS FEM pipeline, this work introduces spatially varying mechanical properties that more realistically

represent the natural variation in breast tissue composition observed in clinical studies. The primary focus is on how these property variations affect external breast surface motion patterns during dynamic loading, as these surface deformations are what optical imaging systems observe.

Building on the finite element framework previously developed within the EWS project by Wielstra (2024), which provided a validated homogeneous breast model for studying fundamental breast mechanics, the present study extends this framework by introducing spatially varying material properties. This heterogeneous implementation enables a more realistic representation of tissue property distributions while preserving continuity with the established baseline model.

The primary focus of this work is on how these property variations affect external breast surface motion patterns during dynamic loading, as these surface deformations are what optical imaging systems observe.

The central research question guiding this work is: How does incorporating tissue heterogeneity and glandular lobular structure affect the biomechanical response of the breast model during simulated motion, and how do these changes manifest in observable surface displacement patterns? This study addresses this question through the implementation and analysis of spatially varying material properties in both adipose and glandular tissue compartments.

Material and method

Overview

This study builds on the Early Warning Scan FEM pipeline to improve a breast model to a more realistic biomechanical model for radiation free cancer detection. The primary contribution is the implementation of spatial heterogeneity in material properties and a different glandular form of distribution to more realistically represent the natural variation in breast tissue as observed in clinical studies. The methodology encompasses model geometry, mesh generation, material property definition, heterogeneous tissue implementation, glandular lobules implementation, and post process analysis. All simulations were performed using FEBio version 4.0 (Maas et al., 2012), a nonlinear finite element solver specifically designed for biomechanical applications. Model generation and analysis were performed using Python 3.13, with Blender used for visualization. This section first describes the existing computational framework and simulation workflow, then details the heterogeneity implementation implemented in this study.

Computational Environment and Software

The computational pipeline was implemented using Python 3.13 as the primary programming language for pipeline orchestration and data processing. Automated 3D tetrahedral mesh generation was performed using Gmsh 4.14 (Geuzaine & Remacle, 2009), an open-source finite

element mesh generator suited for biological geometries. The finite element solver FEBio 4.0 (Maas et al., 2012) handled all simulations, leveraging specialized capabilities for soft tissue mechanics including large deformations and time dependent material behavior.

Supporting computational tools included NumPy for numerical array operations and linear algebra, Pydantic for configuration file validation and type checking to ensure data integrity, meshio for reading and parsing VTK (Visualization Toolkit) files, a standard format for storing 3D mesh and simulation data, from FEBio simulations and Matplotlib for post processing visualization of simulation results. All simulations were executed on a MacBook Air 2019, with Intel core i5, 8GB of RAM, running macOS Sonoma 14.7.8.

The computational framework consists of multiple interconnected Python modules that manage the complete simulation workflow. Key modules include: `generate_mesh.py` for automated mesh generation using the Gmsh Python Api, `model_settings.py` defining geometric parameters and mesh specifications using Pydantic data models, `simulation_settings.py` specifying material properties, boundary conditions, and solver parameters, `write_to_feb.py` for assembling and exporting FEBio input files in XML format, `data_analysis.py` for post processing VTK output files and generating analysis visualizations.

Geometric Model and Finite Element

The model of a breast represents a simplified hemisphere with posterior chest wall attachment. This geometric approximation captures the essential features necessary for analyzing tissue deformation under gravitational and small movement. The existing pipeline uses an axisymmetric approach where a 2D cross section is revolved 360° around the vertical axis to create the 3D model. While this geometric simplification ensures computational efficiency and was suitable for initial model development, this paper's work extends the model's realism by introducing spatial heterogeneity in material properties.

The geometry is constructed as a 2D cross section (Figure 1) which is subsequently revolved around the Y-axis. This approach ensures geometric consistency while significantly reducing the complexity of initial model construction, as only the planar profile needs to be defined before automatic revolution creates the full three-dimensional structure. Note that Figure 1 shows the cross section in a demonstrative orientation for clarity; the actual geometric construction in the code revolves around the vertical Y-axis.

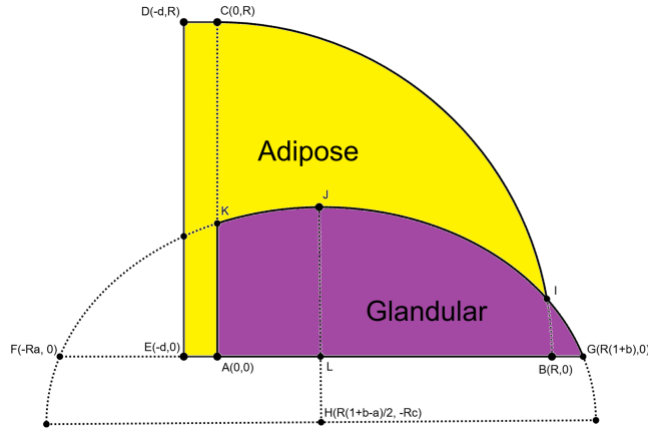


Figure 1: Two-dimensional geometry of the breast model showing adipose tissue (yellow), glandular tissue (purple), and key geometric control points.

The model incorporates several key dimensional parameters, that were established in the original pipeline design. The breast radius measures 70 mm, representing a medium sized breast appropriate for the study population and consistent with data from biomechanics literature (Gefen & Dilmoney, 2007). A chest wall thickness of 2 mm provides a computational boundary region where kinematic constraints can be applied, preventing numerical artifacts in the stress field within the breast tissue itself. The 2D geometry is constructed from a series of control points (labelled A through L in Figure 1) that define the tissue boundaries. Starting from origin point A at the chest wall, a circular arc forms the outer breast contour extending to the superior apex at C and the lateral extent at B. An elliptical curve delineates the glandular adipose interface, with its shape controlled by three dimensionless parameters (a , b , c) that determine the ellipse position and curvature to create anatomically realistic nipple geometry. A thin rectangular region posterior to the breast (points A, C, D, E) represents the chest wall interface. This 2D cross section is then revolved 360° around the vertical Y-axis to generate the full three-dimensional axisymmetric breast model, automatically creating the tetrahedral volume elements during the revolution process.

The model distinguishes four distinct tissue regions: adipose tissue comprising the majority of the breast volume, an elliptical glandular region in the central breast area, a thin skin shell, and a chest wall interface surface. This tissue compartmentalization allows for assignment of distinct mechanical properties to each anatomical region while maintaining continuity at tissue interfaces through conformal meshing.

The finite element discretization employs second-order tetrahedral elements, designated as tet10 elements due to their characteristic ten nodes per element (four corner nodes and six mid-edge nodes). This element type was adopted following the original EWS finite element pipeline developed by Wielstra (2024), where it was shown to provide a robust and accurate discretization for breast tissue mechanics.

Tet10 elements demonstrate improved accuracy in representing curved geometries and smooth deformation compared to their linear counterparts. The quadratic shape functions allow

displacement to vary smoothly and nonlinearly within each element, yielding a better approximation of continuous deformation fields and reducing numerical stiffening effects in soft tissue simulations.

Second, these higher order elements exhibit reduced susceptibility to volumetric locking. The additional degrees of freedom at mid-side nodes help accommodate the incompressibility constraint without over constraining element deformation. Third, the mid-side nodes in tet10 elements enable more accurate representation of stress gradients within each element, particularly near tissue interfaces where changes in material properties can create regions of elevated stress.

The mesh generation process utilizes a characteristic element size of 5 mm as the base length scale, which is then refined adaptively based on local geometric features through Gmsh's build-in size field algorithms. The mesh density parameter is set to 100 elements per unit length for testing and development, resulting in finer resolution in regions with high curvature (such as near the nipple and at tissue interfaces) while maintaining computational efficiency in bulk tissue regions where the geometry varies smoothly.

For the default breast geometry, this adaptive approach produces 29121 tetrahedral elements and 35173 nodes, providing sufficient resolution to capture stress gradients without using excessive computational cost. High order optimization algorithms are applied during mesh generation to improve element quality, adjusting mid-side node positions to minimize element distortion while preserving the underlying geometric features. This optimization is particularly important for higher order elements where poorly positioned mid-side nodes can lead to negative Jacobian (severely distorted elements) and simulation failure.

Technical considerations regarding element node ordering conventions between Gmsh and FEBio are handled automatically during mesh export in the `simulation_settings.py` module, ensuring compatibility between the two software packages.

Material properties and constitutive modelling

The existing pipeline implements the Mooney-Rivlin hyper elastic constitutive model for all tissue types. This phenomenological model captures the nonlinear stress-strain behavior characteristic of soft biological tissues through a strain energy density function. The Mooney-Rivlin model is defined by a strain energy density function $W = c_1(I_1 - 3) + c_2(I_2 - 3) + (\kappa/2)(J - 1)^2$, where I_1 and I_2 are the first and second strain invariants that mathematically describe tissue deformation (I_1 characterizes extension/compression, I_2 characterizes shear), where c_1 and c_2 are material coefficients governing the nonlinear shear stiffness of the tissue, with their sum ($c_1 + c_2$) approximating the small-strain shear modulus, κ is the bulk modulus enforcing near incompressibility, and J represents the volume ratio. This model balances computational efficiency with the ability to capture essential nonlinear behaviour of soft tissues, including strain stiffening under large deformations (Galbreath & Goswami, 2025).

The baseline material parameters implemented in the original pipeline represent homogeneous tissue properties, meaning they are constant throughout each tissue compartment without spatial variation. For adipose tissue, the shear parameters c_1 and c_2 are set to 109 Pa and 106 Pa, with a bulk modulus of 425,000 Pa and density of 911 kg/m^3 . Glandular tissue exhibits greater stiffness with $c_1 = 230 \text{ Pa}$ and $c_2 = 195 \text{ Pa}$, reflecting the presence of fibrous connective tissue and increased cellular density, while maintaining the same bulk modulus and a higher density of $1,041 \text{ kg/m}^3$. The skin is modeled as considerably stiffer with both shear parameters set to 1,200 Pa, capturing the contribution of collagen and elastin fibers in the dermis, along with a bulk modulus of 480,000 Pa and density of $1,100 \text{ kg/m}^3$. These parameter values are consistent with experimental measurements reported by Samani et al. (2007) for breast tissue mechanical properties obtained through ex vivo compression testing.

Simulation Workflow and Boundary Conditions

The simulation pipeline follows six sequential stages Figure 2 (flowchart of the pipeline).

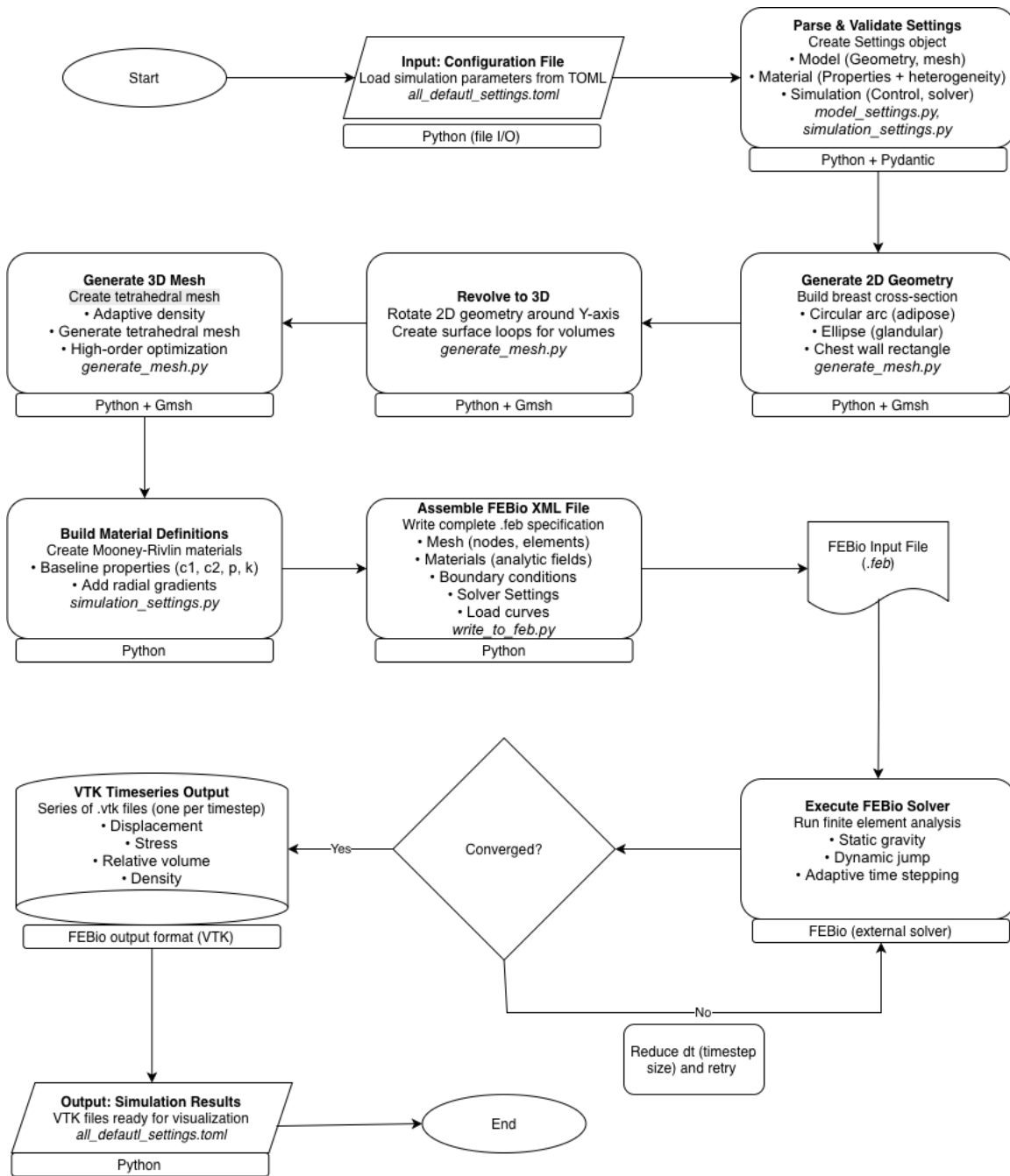


Figure 2: Computational pipeline workflow showing six sequential stages: (1) Configuration validation (Python/Pydantic), (2) 2D mesh generation (Gmsh Python API), (3) 3D mesh generation via revolution (Gmsh), (4) FEBio file assembly (Python), (5) FEBio simulation execution, and (6) post-processing and visualization (Python/meshio/VTK).

The process begins with configuration and validation, where input parameters are read from a TOML configuration file, a human readable text format (for example `gradient_lobules.toml`) containing geometric dimensions, material properties including heterogeneity parameters, mesh settings, and solver configuration. The Pydantic library validates all inputs against predefined schemas defined in the Settings class, ensuring type correctness and parameter ranges before any

computationally expensive operations begin. This validation step catches common errors such as negative material parameters, impossible geometric configurations, or incompatible solver settings.

In the second stage, the `generate_mesh.py` module constructs the breast cross-section using Gmsh's Python API. Key geometric points are positioned based on validated parameters, curves connecting these points define tissue boundaries, surface loops enclose adipose and glandular regions, and distinct material tags are assigned to enable subsequent material property assignment. The 2D mesh consists of triangular elements that will form the faces of tetrahedral volume elements after revolution.

The third stage performs the 3D mesh generation by revolving the 2D profile 360° around the vertical Y-axis. Gmsh automatically creates tetrahedral volume elements (tet10) during this revolution process, applying adaptive density based on local curvature to ensure adequate resolution of geometric features. The high-order optimizer then adjusts node positions to improve element quality metrics. Finally, surface loops are identified and tagged for subsequent application of boundary conditions, including the anterior skin surface, the chest wall surface, and the internal tissue interfaces.

Material definition and FEBio file assembly constitute the fourth stage. The `write_to_feb.py` module extracts node coordinates and element connectivity from the Gmsh mesh data structure, converting from Gmsh's internal format to FEBio's expected node and element numbering. Material definitions are written with math expressions for heterogeneous properties, allowing FEBio to evaluate spatially varying material parameters at each integration point during the simulation. Boundary conditions are defined for the fixed chest wall and parabolic displacement prescription, solver parameters are configured including convergence tolerances and time-stepping options, and output requests are specified to control which field variables (displacement, stress, strain, etc.) are written to result files. The complete specification is then written to a FEBio XML input file (.feb) with proper formatting and structure validation.

The fifth stage executes the FEBio simulation. This launches FEBio as a subprocess, redirecting standard output and error streams to log files for later analysis. The simulation proceeds in two sequential steps, each with its own analysis type, time discretization, and loading conditions.

Step 1 implements gravity preload over a duration of 0 to 1 second, divided into 10 sub steps. During this phase, the chest wall surface is fully fixed, while a body force representing gravitational acceleration ($pg = 9.81 \text{ m/s}^2$) is ramped linearly from 0 to 100% of its full magnitude. This preconditioning step establishes physiologically relevant pre-stress in the tissue before dynamic loading begins, ensuring that the initial stress state reflects the gravitational loading present during normal breast motion rather than starting from an unrealistic zero-stress state.

Step 2 implements dynamic parabolic motion from 1 to 2.2 seconds (total duration 1.2 seconds), divided into 120 time steps with adaptive time stepping enabled. The chest wall surface experiences prescribed vertical displacement following a parabolic trajectory derived from

projectile motion equations. The vertical displacement follows $z(t) = v_0(t-1) - \frac{1}{2}g(t-1)^2$ where $v_0 = \sqrt{2gh_{\text{max}}}$ represents the initial vertical velocity required to reach maximum height h_{max} . The default maximum height is set to 1 cm; this yields an initial velocity of $v_0 = 0.443$ m/s and a time in the air of 0.09 seconds. Horizontal displacements (X and Y directions) are constrained to zero during this phase, restricting motion to purely vertical displacement of the chest wall. Mass proportional damping is applied through the damping matrix $C = \beta_m \cdot M$, where the mass damping coefficient $\beta_m = 20 \text{ s}^{-1}$ and M is the global mass matrix (with units of kg). The resulting damping matrix C has units of kg/s, providing energy dissipation during breast oscillations and preventing unrealistic sustained vibrations. This damping coefficient was inherited from the existing simulation pipeline and achieves oscillation decay over 1 second following the peak displacement

FEBio's default nonlinear solver employs a Newton-Raphson method with BFGS (Broyden-Fletcher-Goldfarb-Shanno) quasi-Newton updates to approximate the tangent stiffness matrix, updating this approximation at each iteration based on the change in residual forces. Convergence criteria check, with a maximum of 15 iterations per time step before declaring convergence failure. An automatic time-stepping algorithm adjusts step sizes based on convergence behavior, with a maximum time step of $dt_{\text{max}} = 0.01$ s during Step 2, a cutback factor of 0.5 (halving the time step upon convergence failure), and a maximum of 20 retries with progressively smaller time steps before abandoning the solution. This adaptive strategy ensures robust convergence even during rapid changes in boundary conditions, such as at the initiation of the parabolic jump when accelerations are highest.

Output data is recorded only during Step 2 (the dynamic jump phase). Step 1 serves solely to establish gravitational equilibrium and is not written to VTK output files. VTK output begins at $t = 1.025$ s, which represents the gravity equilibrium state shortly after Step 1 completion and marks the effective initiation of the parabolic jump analysis in Step 2. This means all reported results use the gravity equilibrated configuration as the baseline reference state, with $t = 1.025$ s corresponding to VTK time step 0. Subsequent output is written at regular intervals throughout the jump motion ($t = 1.025$ s to 2.2s), capturing the transient response, oscillation dynamics, and return to steady state.

The sixth and final stage processes simulation output through the `data_analysis.py` module. VTK files generated at each output time point are read using `meshio`, extracting nodal displacement vectors, element wise Cauchy stress tensors, relative volume changes (J values), and material identification tags. Von Mises equivalent stress is computed from the Cauchy stress tensor as:

$$\sigma_{VM} = \sqrt{\frac{1}{2}[(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\sigma_{xy}^2 + \sigma_{yz}^2 + \sigma_{xz}^2)]}$$

This scalar metric quantifies the magnitude of shape changing (deviatoric) stress by combining all six independent components of the stress tensor. The stress tensor describes internal forces within the material through three normal stresses ($\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$), which act along the x, y, and z directions and represent tension or compression, and three shear stresses ($\sigma_{xy}, \sigma_{yz}, \sigma_{xz}$), which act

diagonally and cause layers of material to slide past each other. The von Mises stress condenses these components into a single value that represents the effective stress driving material yielding and is widely used in biomechanics to assess tissue loading.

For displacement analysis, two approaches were used: landmark displacement at five anatomical positions (nipple, left/right lateral, superior, and inferior) is reported as relative displacement from the gravity equilibrium baseline, whereas surface displacement analysis uses absolute displacement from the original undeformed configuration.

Statistical summaries are computed for each time step, including mean, median, and maximum von Mises stress across all elements and tissue specific statistics separating adipose and glandular tissue responses. Visualization routines generate time series plots of stress evolution, multi-model stress comparison plots, tissue specific analysis separating von Mises stress and volume change by tissue type, spatial displacement maps showing surface displacement patterns in top view for all model variants, surface displacement evolution over time, and landmark displacement comparisons across models.

Spatial Heterogeneity

Real breast tissue exhibits spatial variation in mechanical properties due to natural biological variability, gradual transitions between tissue types rather than sharp boundaries, and regional density differences. MRI elastography studies have demonstrated substantial variability in breast tissue stiffness, with fibroglandular tissue exhibiting coefficients of variation of approximately 48% (7.5 ± 3.6 kPa) and adipose tissue exhibiting 58% variability (3.3 ± 1.9 kPa) across healthy volunteers (McKnight et al., 2002; Patel et al., 2020). Mechanical testing of ex vivo human breast tissue has confirmed this heterogeneity, with measured standard errors of approximately 20% within tissue samples (Goodbrake et al., 2022). Additionally, glandular tissue demonstrates approximately 2-3 times the stiffness of adipose tissue, with this ratio varying spatially throughout the breast (Patel et al., 2020). The baseline homogeneous model in the existing pipeline does not capture this heterogeneity.

The primary goal of implementing spatial heterogeneity was to investigate how spatial tissue property variations affect the external breast surface motion patterns during dynamic loading, as these surface deformations are what optical imaging systems observe. Secondary objectives include examining how property gradients influence internal stress distributions and whether localized property variations create detectable changes in surface displacement fields.

Model configurations

Eight model configurations were developed to systematically evaluate heterogeneity effects (Table 1). The baseline model represents fully homogeneous tissue without spatial variation, where adipose and glandular tissues maintain distinct material properties (adipose: $\rho = 911 \text{ kg/m}^3$, $c_1 =$

109 Pa, $c_2 = 106$ Pa; glandular: $\rho = 1041$ kg/m³, $c_1 = 230$ Pa, $c_2 = 195$ Pa) but without any spatial gradients within each tissue compartment.

Table 1: Material parameter configurations for the eight model variants tested. ρ = density, c_1 and c_2 = Mooney-Rivlin shear parameters, K = bulk modulus (constant at 425,000 Pa for adipose and glandular tissue). Radial heterogeneity parameters (a) control the amplitude of spatial variation: $a = 50$ for density, $a = 30$ for c_1 , and $a = 25$ for c_2 in adipose tissue. Lobular heterogeneity uses Gaussian structures with amplitudes: 35 kg/m³ (density), 70 Pa (c_1), and 55 Pa (c_2) in glandular tissue. All models use identical skin parameters ($\rho = 1100$ kg/m³, $c_1 = c_2 = 1200$ Pa, $K = 480,000$ Pa).

Model	Adipose ρ (kg/m ³)	Adipose c_1 (Pa)	Adipose c_2 (Pa)	Glandular ρ (kg/m ³)	Glandular c_1 (Pa)	Glandular c_2 (Pa)	Bulk Modulus K (Pa)
Baseline	911	109	106	1041	230	195	425,000
Radial gradient (density only)	911 + gradient ($a=50$)	109	106	911	230	195	425,000
Radial gradient (coef only)	911	109 + gradient ($a=30$)	106 + gradient ($a=25$)	911	230	195	425,000
Lobular (Glandular only)	911	109	106	911 + 6 lobules ($a=35$)	230 + 6 lobules ($a=70$)	195 + 6 lobules ($a=55$)	425,000
Full model (Radial + Lobular)	911 + gradient ($a=50$)	109 + gradient ($a=30$)	106 + gradient ($a=25$)	911 + 6 lobules ($a=35$)	230 + 6 lobules ($a=70$)	195 + 6 lobules ($a=55$)	425,000
Adipose-only density	911	109	106	911	230	195	425,000
Glandular-only density	1079	109	106	1079	230	195	425,000
Complete Homogeneous	911	109	106	911	109	106	425,000

Intermediate models isolate specific heterogeneity components to understand their individual contributions. Radial gradient (density only) varies adipose density while maintaining constant stiffness, implementing a continuous gradient from the chest wall toward the breast surface. Radial gradient (coefficients only) varies Mooney-Rivlin parameters c_1 and c_2 in adipose tissue while keeping density constant. Lobular (glandular only) adds six Gaussian lobules to glandular tissue simulating fibroglandular lobular architecture. The full model combines radial gradients in adipose tissue with lobular structures in glandular tissue, representing the most comprehensive heterogeneity implementation.

Three additional control configurations test the effects of eliminating tissue type differentiation. Adipose-only density assigns adipose tissue density (911 kg/m³) to both adipose and glandular compartments while maintaining tissue specific stiffness parameters (adipose: $c_1 = 109$ Pa, $c_2 =$

106 Pa; glandular: $c_1 = 230$ Pa, $c_2 = 195$ Pa), isolating the mechanical effects of density differences. Glandular-only density assigns glandular tissue density (1079 kg/m^3) to both compartments while maintaining tissue specific stiffness parameters, providing a complementary test of density effects. Complete homogeneous eliminates all tissue differentiation by assigning adipose material properties ($\rho = 911 \text{ kg/m}^3$, $c_1 = 109$ Pa, $c_2 = 106$ Pa) to both compartments, creating a truly uniform material throughout the breast volume. These three homogeneous models provide reference points to quantify how much mechanical behaviour depends on tissue type differentiation (in the baseline model) versus spatial heterogeneity within tissue compartments (in the gradient and lobular models).

All models maintain $K = 425,000$ Pa for adipose and glandular tissue, and constant skin properties ($\rho = 1100 \text{ kg/m}^3$, $c_1 = c_2 = 1200$ Pa, $K = 480,000$ Pa).

Implementation of Heterogeneity Patterns

Two patterns of heterogeneity were implemented to represent different aspects of breast tissue architecture. Radial heterogeneity gradients were applied to the adipose compartment, which represents the largest tissue volume and therefore has the greatest potential impact on global mechanical behavior. Lobular heterogeneity patterns model the discrete glandular lobule structures, representing the natural lobular architecture of breast tissue.

Both patterns use analytic mathematical functions evaluated by FEBio at integration points during the simulation, rather than discrete nodal values assigned during preprocessing. This approach ensures smooth, continuous property fields without numerical artifacts that can arise from discontinuous property assignments, allows flexible modification of heterogeneity patterns without remeshing since the mathematical expressions can be changed in the input file, and reduces memory requirements as no per-element property storage is needed. The heterogeneity is applied to all three material parameters: c_1 , c_2 , and ρ (density).

Radial Heterogeneity Gradient

The radial heterogeneity gradient implements a continuous increase in material properties from the chest wall toward the breast surface in the adipose compartment, motivated by the anatomical observation that adipose tissue near the breast surface often contains increased fibrous content due to Cooper's ligament integration and superficial fascial attachments (Gefen & Dilmoney, 2007).

The mathematical formulation takes the form $P(x, y, z) = P_0 + \alpha_p(1 - \exp(-((x - xc)^2 + (y - yc)^2 + (z - zc)^2)/L^2))$, where (xc, yc, zc) represents the gradient center, positioned at the chest wall origin in this project. The parameter α_p controls the maximum property increase far from the centre, with units matching the property being modified (Pa for Mooney-Rivlin coefficients c_1 and c_2 , or kg/m^3 for density ρ), while L sets the characteristic length scale that governs how quickly the

gradient develops. The exponential form ensures smooth, continuous property transitions and radial symmetry (properties depend only on distance from the starting point), with L controlling the spatial extent of the gradient.

The specific parameter values for the radial gradient were selected through iterative testing. The gradient center was positioned at $(x_c, y_c, z_c) = (0, 0, 0)$ mm at the chest wall origin, establishing the reference point from which distance is measured and ensuring the gradient increases radially outward from the attachment point. The characteristic length L was set to 55 mm to ensure the gradient develops over most of the tissue volume while maintaining smooth transitions. At distance $r = L$ from the center, the exponential term $\exp(-r^2/L^2)$ equals $\exp(-1) \approx 0.37$, meaning the property value reaches approximately 63% of its maximum change ($1 - 0.37 = 0.63$). This creates a gradual gradient that avoids numerical artifacts from steep property changes while generating measurable mechanical differences between posterior and anterior regions.

Stiffness Gradient Parameters (c_1, c_2)

Gradient amplitudes for the Mooney-Rivlin coefficients were set as $\alpha_{c_1} = +30$ Pa and $\alpha_{c_2} = +25$ Pa, representing 28% and 24% increases over baseline adipose values ($c_1 = 109$ Pa, $c_2 = 106$ Pa), respectively. These modest gradients model within adipose tissue variation due to varying fat composition and fibrous content. The amplitude ratio $\alpha_{c_1}/\alpha_{c_2} = 1.20$ was chosen to be in the same range as the baseline ratio $c_1/c_2 = 1.03$, preserving similar contributions of the first and second strain invariants to the strain energy function.

Density Gradient Parameter (ρ)

The density gradient amplitude was set as $\alpha_{\rho} = +50$ kg/m³ (5.5% increase over baseline adipose density of 911 kg/m³). This small variation reflects physiological constraints: while the density difference between adipose and glandular tissue is only 22% (900 vs 1,100 kg/m³; Griffiths et al., 2009), the model uses a more conservative 14% difference (911 vs 1,041 kg/m³) because the gradient represents partial fibrous infiltration rather than complete tissue replacement. Larger density gradients were avoided to prevent inertial effects from dominating the mechanical response during dynamic loading, which would obscure the intended investigation of stiffness gradient effects on surface motion.

Initial stiffness estimates were based on literature noting that glandular tissue exhibits approximately 2–3 times the stiffness of adipose tissue (Sinkus et al., 2007). Rather than implementing gradients strong enough to represent a full transition from adipose to glandular properties, the goal was to apply modest gradients reflecting variation within adipose tissue due to fat composition and fibrous content. Target increases of 25–30% were chosen to remain physiologically plausible while producing measurable mechanical effects.

Lobular Heterogeneity Pattern

The lobular heterogeneity pattern models localized regions of varying stiffness using Gaussian (bell shaped) distributions centered at specified points in space, representing the natural lobular architecture of glandular breast tissue. The mathematical formulation takes the form:

$$P(x,y,z) = P_0 + \sum_i A_i \exp(-((x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2)/\sigma_i^2)$$

Where P represents any material property (c_1 , c_2 , or ρ), P_0 is the baseline homogeneous value for that property, and the summation $\sum_i$ adds contributions from all lobules. For each lobule i , (x_i, y_i, z_i) specifies its center coordinates, A_i is the amplitude controlling the magnitude of property variation (positive values create stiffer regions, negative values create softer regions), and σ_i controls the spatial extent of the variation analogous to the standard deviation of a Gaussian distribution.

Gaussian distributions were chosen for their smooth, continuous transitions and ability to superpose multiple lobules without numerical discontinuities.

For the glandular tissue, six lobular regions were introduced to mimic localized heterogeneity within the glandular tissue. Each lobule was modeled as a Gaussian shaped variation in material properties, defined by its center coordinates, width, and amplitude. The centres were positioned symmetrically around the glandular region midpoint at (2.0, 4.8, 0.0), (1.0, 4.8, 1.7), (-1.0, 4.8, 1.7), (-2.0, 4.8, 0.0), (-1.0, 4.8, -1.7), (1.0, 4.8, -1.7) in cm, forming a hexagonal pattern in the horizontal plane at a depth of 4.8 cm from the chest wall. Each lobule had a width of 5 mm and amplitude parameter of approximately ± 70 Pa and ± 55 Pa for shear moduli, and ± 35 Pa for density variations, resulting in stiffness changes of $\pm 30\%$ for c_1 and for $\pm 28\%$ c_2 relative to baseline glandular properties. This configuration creates spatially localized regions of increased or decreased stiffness, enabling assessment of how heterogeneity influences tissue mechanics.

These amplitudes should be chosen to remain within physiologically plausible ranges based on the coefficients of variation (20-40%) reported for breast tissue properties (Samani et al., 2007), create measurable stress perturbations that could reveal the influence of heterogeneity on surface displacement patterns, and test the model's sensitivity to localized property changes that might represent natural tissue variation.

The implementation defines heterogeneity parameters in the Heterogeneity class (containing Lobule objects and radial gradient parameters). These are converted to mathematical expressions by the `_expr_with_hetero()` method in MaterialProperties, which builds analytic formulas evaluated by FEBio at each integration point during simulation.

For example, the c_1 parameter with a single lobule might be exported as:

```
<c1 type="math">230+25*exp(-((X-0.035)^2+(Y-0.040)^2+(Z-0)^2)/(0.01^2))</c1>
```

FEBio evaluates this expression at each Gauss integration point within each element during the simulation, allowing property values to vary continuously in space according to the specified mathematical function.

Combined Heterogeneity Patterns

When both lobular and radial patterns are implemented, each tissue type has its own heterogeneity appropriate to its anatomical structure. Within each compartment, the final property field is computed as $P(x,y,z) = P_0 + P_{\text{radial}}(x,y,z) + P_{\text{lobular}}(x,y,z)$, where $P_{\text{radial}}(x,y,z)$ represents the radial gradient (applied to adipose tissue) and $P_{\text{lobular}}(x,y,z)$ represents the sum of Gaussian lobular terms (applied to glandular tissue).

This approach allows the radial gradient to capture the posterior to anterior tissue density trend in adipose, while lobular variations in glandular tissue represent the discrete lobular architecture characteristic of that tissue type. Independent control of each component enables systematic investigation of their respective influences on mechanical behaviour and surface motion patterns through simulation studies where one component is varied while holding the other constant.

Verification of Implementation

Verification of correct implementation proceeded through several checks. Syntax validation confirmed that FEBio successfully parsed all mathematical expressions without errors, as indicated by normal termination messages in the solver log files rather than parsing errors or premature exits.

A range check involved extracting density values from the VTK output files and computing minimum and maximum values across all nodes, confirming that the minimum density remained near 911 kg/m^3 (baseline adipose) in regions distant from both the chest wall and any lobules, while the maximum density reached approximately 960 kg/m^3 in regions where the radial gradient contributes its full amplitude.

Results

The finite element simulations successfully captured breast tissue deformation under gravitational loading and dynamic motion across all eight model configurations. Results are presented in three categories: surface displacement patterns, which describe external deformation visible to optical imaging systems; stress distributions, which reveal internal mechanical loading states; and temporal dynamics, which track the evolution of mechanical responses during the parabolic jump motion. All analyses compare heterogeneous model variants with the baseline homogeneous configuration to isolate the effects of material property variations.

Surface Displacement Patterns

Peak Vertical Displacement

Peak Vertical displacement at the breast surface varied across models, ranging from 12.2 mm to 20.1 mm, representing a 65% variation range (Table 2). The baseline homogeneous model produced a maximum displacement of 14.0 mm.

Table 2: Peak vertical displacement at the breast surface for each model configuration during the parabolic jump simulation (step 3, $t \approx 1.075$ s). Values represent maximum displacement in the vertical direction relative to the gravity equilibrium state. The baseline homogeneous model (14.0 mm) serves as the reference for comparison.

Model	Vertical displacement (mm)
Radial gradient (density only)	13.7
Radial gradient (coef only)	12.2
Lobular (Glandular only)	15.3
Full model (Radial + Lobular)	13.4
Adipose-only density	15.4
Glandular-only density	20.1
Complete Homogeneous	17.2

Stiffness gradients (c_1 , c_2) reduced displacement by 13% (from 14.0 mm to 12.2 mm), representing the strongest displacement constraining effect among heterogeneous variants. In contrast, density gradients alone produced minimal change (-2%, to 13.7 mm). The Full Model, combining both radial stiffness and density gradients with lobular structures, showed intermediate displacement (13.4 mm, -4% from Baseline), positioned between the Radial coef model (12.2 mm) and the Radial density model (13.7 mm).

Models with exclusively density-based heterogeneity (Glandular-only density, Adipose-only density, Lobular) showed increased displacement relative to Baseline (+9% to +44%), while models incorporating stiffness gradients showed reductions (-2% to -13%). The Glandular-only density model exhibited the largest displacement (20.1 mm, +44%), while the Complete Homogeneous model showed moderate increase (17.2 mm, +23%).

Landmark specific Displacement Analysis

Analysis of five anatomical landmarks at step 3 revealed position dependent displacement patterns across models, with variability ranging from 10% at lateral positions to 89% at the nipple (Figure 3, Table 3).

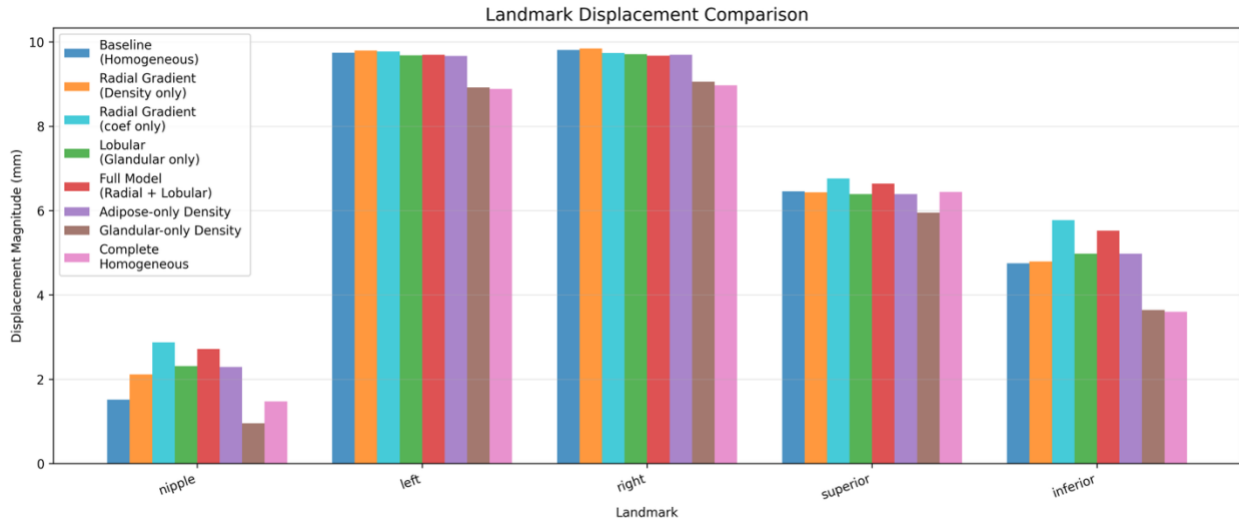


Figure 3: Landmark displacement comparison across eight model configurations at step 3 (peak displacement, $t \approx 1.025$ s). Bar chart enables visual comparison of model-dependent responses at five anatomical positions: nipple (anterior apex), left/right (lateral at $\pm 30^\circ$), superior, inferior. Note the consistent ~ 9 - 10 mm lateral displacement across most heterogeneous models versus the highly variable nipple response.

The visual comparison reveals several patterns. Lateral positions (left/right) show high consistency across heterogeneous models (~ 9 - 10 mm). In contrast, nipple displacement varies substantially (1.0 - 2.9 mm). Baseline consistently shows the smallest displacements except at the posterior position. Exact values are provided in Table 3.

Table 3: Landmark Displacement Comparison at step 3. All values in mm. Radial (p) = Radial Gradient (Density only); Radial (c) = Radial Gradient (coef only). Values represent displacement magnitude at peak dynamic loading (step 3, $t \approx 1.075$ s).

Model	Nipple (mm)	Left (mm)	Right (mm)	Superior (mm)	Inferior (mm)
Baseline	1.52	9.74	9.81	6.46	4.75
Radial (p)	2.11	9.79	9.84	6.43	4.79
Radial (c)	2.88	9.77	9.74	6.76	5.77
Lobular	2.31	9.68	9.71	6.39	4.97
Full Model	2.72	9.69	9.68	6.64	5.52
Adipose p	2.28	9.67	9.70	6.39	4.97
Glandular p	0.96	8.92	9.05	5.95	3.64
Complete Model	1.47	8.88	8.97	6.44	3.60

Landmark Pattern Analysis

Landmark displacement patterns revealed position dependent sensitivity to material heterogeneity (Figure 4). The nipple position exhibited the highest model dependent variability. The baseline model produced 1.52 mm displacement, while heterogeneous models ranged from 2.11 mm (Radial p) to 2.88 mm (Radial c), representing a 39% to 89% increase from the baseline. The radial

stiffness gradient (c only) produced the largest nipple displacement (2.88 mm), indicating this region is sensitive to internal stiffness distributions. This 89% displacement increase from a 28% stiffness gradient demonstrates mechanical amplification effects at this geometrically complex location.

The left and right lateral landmarks showed consistent displacement across all model configurations, ranging from 9.68 to 9.81 mm. The standard deviation across all models was less than 0.06 mm for each lateral position, indicating that material heterogeneity has minimal effect on lateral displacement magnitudes.

The superior landmark showed moderate consistency across models, with displacements ranging from 5.95 to 6.76 mm. The baseline produced 6.46 mm displacement. Most heterogeneous models remained within $\pm 5\%$ of this baseline value, with only the Radial c model showing a notable increase (6.76 mm, +5%) and the Glandular ρ model showing a decrease (5.95 mm, -8%).

The inferior landmark exhibited the most model dependent behaviour other than the nipple positions. The baseline showed 4.75 mm displacement, while heterogeneous models ranged from 3.60 mm to 5.77 mm, a 60% variation range. The Radial c model produced the largest inferior displacement (5.77 mm, +21% from baseline), while the Complete Homogeneous and Glandular ρ models showed the smallest values (3.60 and 3.64 mm respectively, representing 24-25% reductions from baseline).

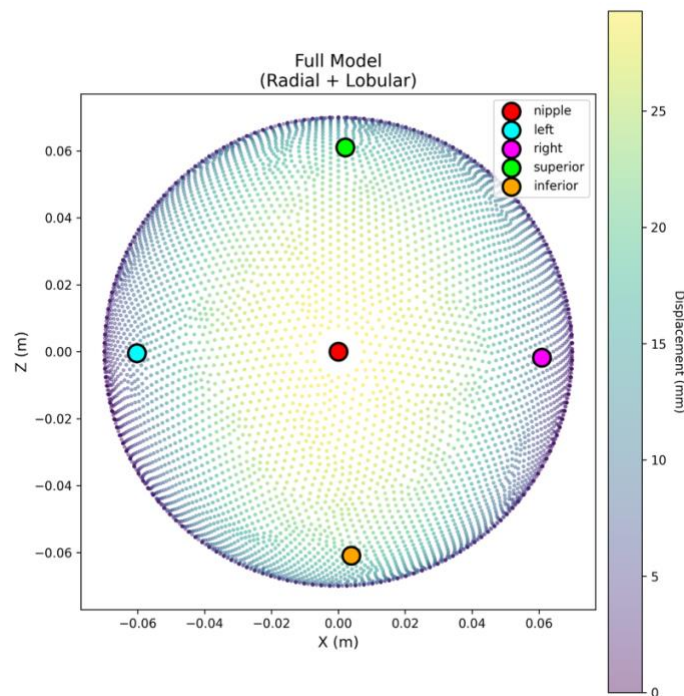


Figure 4: Landmark displacement comparison across eight model configurations at step 3 (peak displacement during parabolic jump). Five anatomical positions: nipple (anterior apex), left/right (lateral at $\pm 30^\circ$), superior (top), inferior (bottom).

Spatial Displacement Maps

The spatial displacement patterns at time step 3 revealed distinct model dependent characteristics (Figure 5). All models displayed radial displacement gradients emanating from the chest wall attachment, with maximum values at the anterior surface (35-45 mm) and minimum values at the posterior boundary (0-10 mm). These spatial patterns reflect the combined effects of gravitational loading and the transient parabolic jump motion.

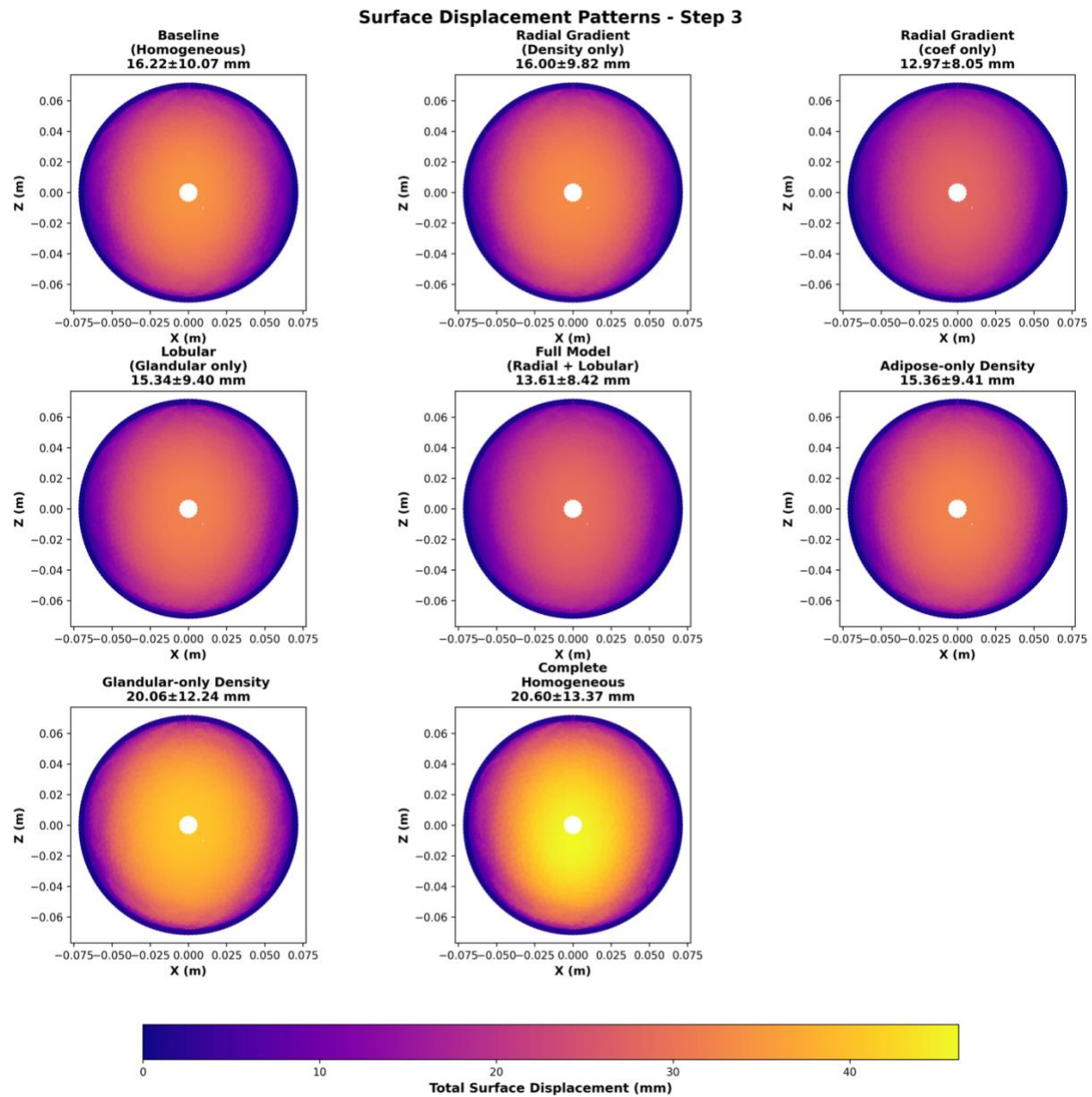


Figure 5: Spatial displacement field visualization at step 3 for eight configurations. All panels show absolute displacement magnitude from the original undeformed configuration ($t = 0$ s, before gravity application). Colour scale represents total displacement (mm): yellow orange (35-45 mm), while purple (0-10 mm). Radial Gradient and Full Model show more spatially constrained high displacement zones compared to Baseline.

The radial Gradient (coef only) model showed the smallest mean displacement with sharper spatial gradients, while the Complete Homogenous and Glandular-only density models showed the largest mean displacement (~20 mm) with more uniform displacement fields (Table 4). The Full Model (13.61 mm) combined radial gradients with subtle local variations from glandular lobules, producing intermediate displacement magnitudes.

Table 4: Mean surface displacement across all surface nodes with standard deviation for each model configuration. Values represent absolute displacement magnitude from the undeformed configuration ($t = 0$ s, before gravity application) at step 3 (peak dynamic loading).

Model	Mean Displacement (mm)
Baseline	16.22 $\pm$ 10.07
Radial gradient (density only)	16.00 $\pm$ 9.82
Radial gradient (coef only)	12.97 $\pm$ 8.05
Lobular (Glandular only)	15.34 $\pm$ 9.40
Full model (Radial + Lobular)	13.61 $\pm$ 8.42
Adipose-only Density	15.36 $\pm$ 9.41
Glandular-only Density	20.06 $\pm$ 12.24
Complete Homogeneous	20.60 $\pm$ 13.37

Quantitatively, mean displacement varied by 59% across models, from 12.97 mm (Radial coef) to 20.60 mm (Complete Homogeneous). Standard deviations ranged from 8.05 mm to 13.37 mm, reflecting spatial variability within each model configuration. Models incorporating stiffness heterogeneity (Radial coef, Full Model) demonstrated 28-39% reductions in mean displacement compared to the Complete Homogeneous model.

Surface Displacement Evolution

Mean surface displacement evolved throughout the parabolic jump phase. The temporal evolution for the Full Model is shown in Figure 6a, with evolution patterns for all other model configurations provided in Appendix A Figures 6b-6i. All displacement values represent absolute displacement from the undeformed state ($t = 0$ s) including both gravitational sag and dynamic jump motion.

At $t = 1.0$ s (gravity equilibrium baseline), mean displacement was 18-20 mm across models. At $t = 1.1$ s (minimum displacement during upward acceleration), mean displacement decreased to 13-16 mm depending on model. Displacement then recovered, with models reaching new equilibrium values by $t = 1.2$ s.

The Baseline and Complete Homogeneous models consistently showed larger displacements throughout the time course compared to models with stiffness heterogeneity. The temporal pattern (initial state, transient decrease, recovery, gradual stabilization) was consistent across all models despite magnitude differences.

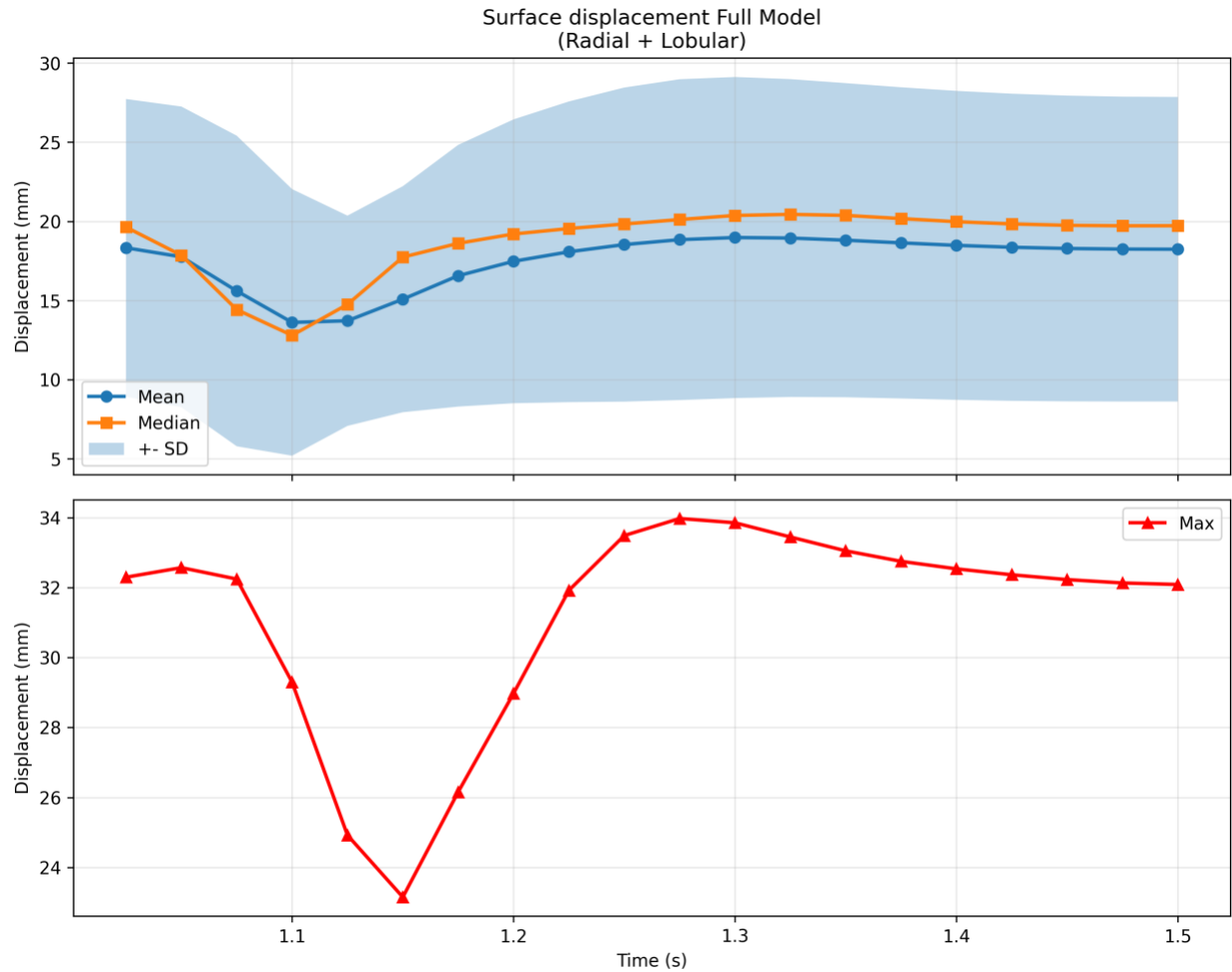


Figure 6a: Surface displacement evolution for Full model. Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three-phase behaviour: (1) gravity equilibrium baseline ($t = 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s). Displacement evolution for other model configurations shown in Appendix A Figures 6b-6h.

Models with stiffness heterogeneity (Radial coef, Full Model) showed smaller displacement magnitudes throughout the time course but similar temporal dynamics. The maximum displacement values (30-38 mm) represent the most mobile surface point (typically near the nipple position) and showed similar three phase behaviour across all model configurations.

Temporal stress Evolution

Transient Response During Jump

Mean von Mises stress evolved through distinct phases during the simulation. At the start of data recording ($t = 1.025$ s, VTK step 0), models showed initial stress levels reflecting gravitational

loading, ranging from 695 Pa to 735 Pa (Table 5).

Table 5: Initial mean von Mises stress at gravity equilibrium. Values represent mean stress across all elements at $t = 1.025$ s before jump initiation (VTK step 0).

Model	Mean stress (Pa)
Baseline	735
Radial gradient (density only)	720
Radial gradient (coef only)	705
Full model (Radial + Lobular)	725
Complete Homogeneous	695

At jump initiation ($t \approx 1.0$ - 1.025 s), stress increased sharply across all models, reaching transient peak values before declining during the upward acceleration phase (Figure 7). Following the transient peak, stress decreased to minimum values around $t = 1.1$ s, with most models reaching 380-470 Pa. The Glandular-only density model exhibited different behaviour, maintaining consistently higher stress levels throughout the simulation. By $t \approx 1.15$ s, stress began recovering as the breast decelerated and reversed direction.

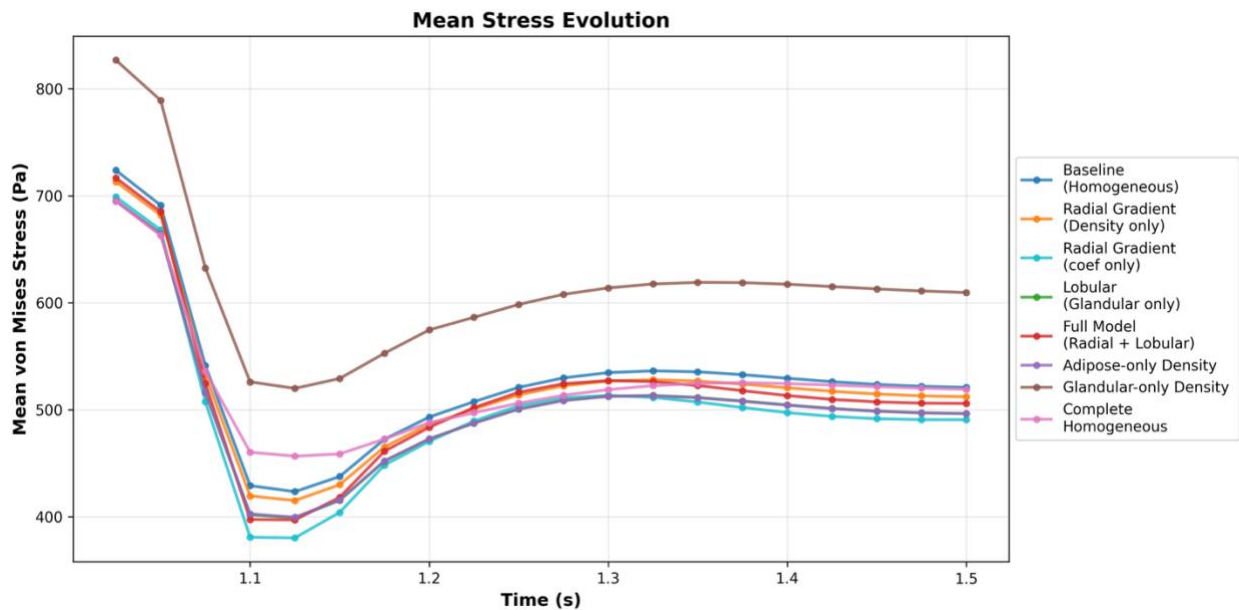


Figure 7: Mean von Mises stress evolution showing transient response from gravity equilibrium state ($t = 1.025$ s baseline). The Glandular Density model (Glandular p) consistently maintains highest stress levels throughout the simulation.

The temporal evolution patterns revealed model dependent stress magnitudes but consistent phase behaviour across configurations. Models with stiffness heterogeneity (Radial coef, Full Model) showed lower initial stress levels compared to the Baseline, while the stress reduction during the acceleration phase was proportionally similar across all models (approximately 45-50%

decrease from initial values).

Steady State Oscillation Behaviour

After the transient jump phase, stress gradually stabilized as oscillations damped. The temporal evolution in Figure 7 shows that all models converged toward steady state conditions by $t \approx 1.3$ - 1.4 s. By $t = 1.5$ s (end of analysed window), steady state mean stresses ranged from 490 Pa to 520 Pa across models (Table 6).

Table 6: Steady state mean von Mises stress at end of simulation window. Values represent mean stress across all elements at $t = 1.5$ s after oscillation damping.

Model	Steady state mean stress (Pa)
Baseline	520
Radial gradient (density only)	515
Radial gradient (coef only)	490
Full model (Radial + Lobular)	510
Complete Homogeneous	518

The Radial gradient (coef only) model showed the lowest steady-state stress (490 Pa, 6% below Baseline), while the Baseline and Complete Homogeneous models exhibited the highest values (520 and 518 Pa respectively). Steady state stress levels were 29-33% lower than initial gravitational equilibrium values (Table 6 vs. Table 5), reflecting the reduced tissue strain after partial recovery from the dynamic loading cycle.

Maximum Stress evolution

Maximum von Mises stress (peak stress across all elements at each time point) exhibited similar temporal patterns to mean stress but with larger magnitudes (Figure 8). Peak maximum stresses occurred at jump initiation ($t \approx 1.025$ s), ranging from 2725 Pa to 3200 Pa across models (Table 7).

Table 7: Peak maximum von Mises stress during parabolic jump. Values represent the maximum stress magnitude across all elements at peak loading (step 3, $t = 1.025$ s).

Model	Peak stress (Pa)
Baseline	3200
Radial gradient (density only)	2740
Radial gradient (coef only)	2730
Full model (Radial + Lobular)	2760
Complete Homogeneous	2725

Heterogeneous models showed 14-15% reductions in peak stress compared to Baseline (from 3200 Pa to 2725-2760 Pa). Following the initial peak, maximum stress decreased to minimum values of approximately 1300-1500 Pa at $t \approx 1.075$ s, then gradually recovered and stabilized at 1750-2150 Pa by $t = 1.5$ s (Figure 8). The ratio of peak to steady state maximum stress ranged from 1.3× to 1.8× across models, indicating sustained stress concentrations even after dynamic loading ceased.

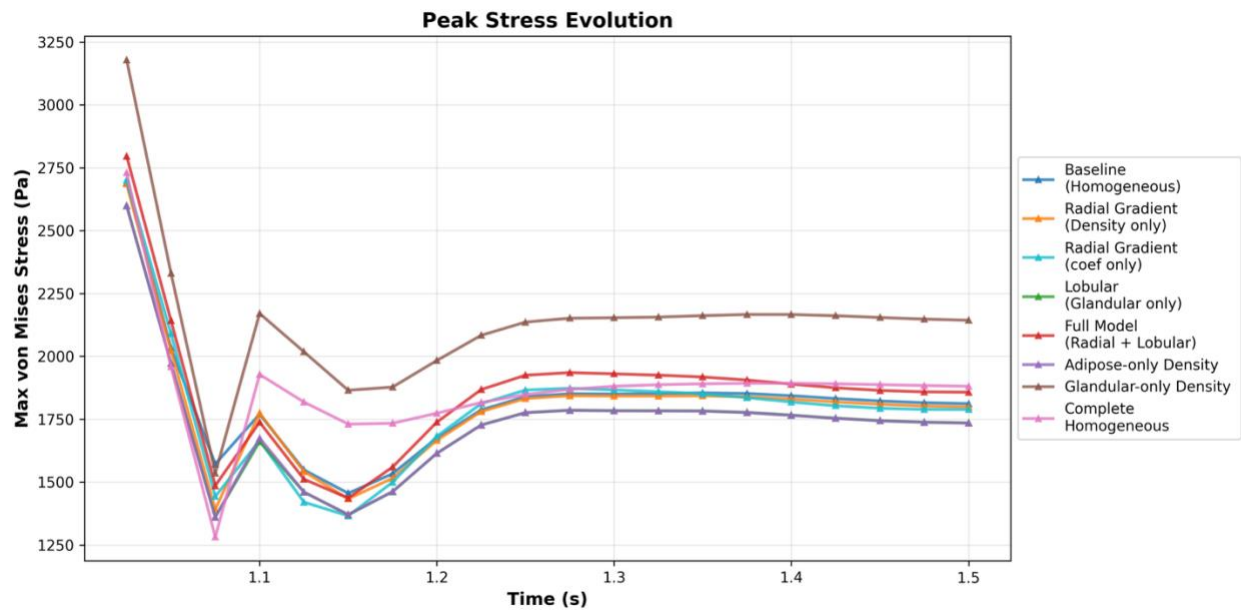


Figure 8: Maximum von Mises stress shows transient response from gravity equilibrium state ($t = 1.025$ s baseline). Glandular p consistently maintains highest stress throughout simulation.

Peak Stress Analysis by Tissue Type

The introduction of spatial heterogeneity altered peak stress magnitudes in both tissue compartments, with tissue specific responses to density and stiffness variations (Figure 9). In glandular tissue, peak von Mises stress ranged from 1.44 kPa to 2.87 kPa across models, representing a 99% variation range (Table 8).

Table 8: Peak von Mises stress in glandular tissue for each model configuration. Values in kPa represent the maximum stress magnitude observed in the glandular compartment during the parabolic jump simulation. The baseline homogeneous model (2.22 kPa) serves as the reference for comparison.

Model	Von Mises Stress (kPa)
Baseline	2.22
Radial gradient (density only)	2.41

Radial gradient (coef only)	2.11
Lobular (Glandular only)	2.35
Full model (Radial + Lobular)	2.79
Adipose-only Density	2.35
Glandular-only Density	2.87
Complete Homogeneous	1.44

Stiffness gradients (coef only) reduced glandular peak stress by 5% relative to Baseline (from 2.22 kPa to 2.11 kPa), while density gradients increased it by 9% (to 2.41 kPa). The Full Model, combining both heterogeneity types, showed a 26% increase (2.79 kPa), while the Glandular-only density model exhibited the highest stress (2.87 kPa, +29%).

In adipose tissue, peak stress ranged from 1.69 kPa to 3.23 kPa, showing a 91% variation range (Table 9). Unlike glandular tissue, both density and stiffness gradients produced similar stress reductions in adipose tissue.

Table 9: Peak von Mises stress in adipose tissue for each model configuration. Values in kPa represent the maximum stress magnitude observed in the adipose compartment during the parabolic jump simulation. The baseline homogeneous model (3.23 kPa) serves as the reference for comparison.

Model	Von Mises Stress (kPa)
Baseline	3.23
Radial gradient (density only)	2.67
Radial gradient (stiffness only)	2.69
Lobular (Glandular only)	2.59
Full model (Radial + Lobular)	2.79
Adipose-only Density	2.59
Glandular-only Density	3.15
Complete Homogeneous	1.69

In adipose tissue, both density and stiffness gradients produced 17-18% reductions in peak stress relative to Baseline (from 3.23 kPa to 2.67-2.69 kPa) (Figure 9). This contrasts with the opposing effects observed in glandular tissue, where density increased stress while stiffness decreased it. The Complete Homogeneous model showed the lowest peak stress in both compartments (1.44 kPa glandular, 1.69 kPa adipose), representing 35-48% reductions from Baseline.

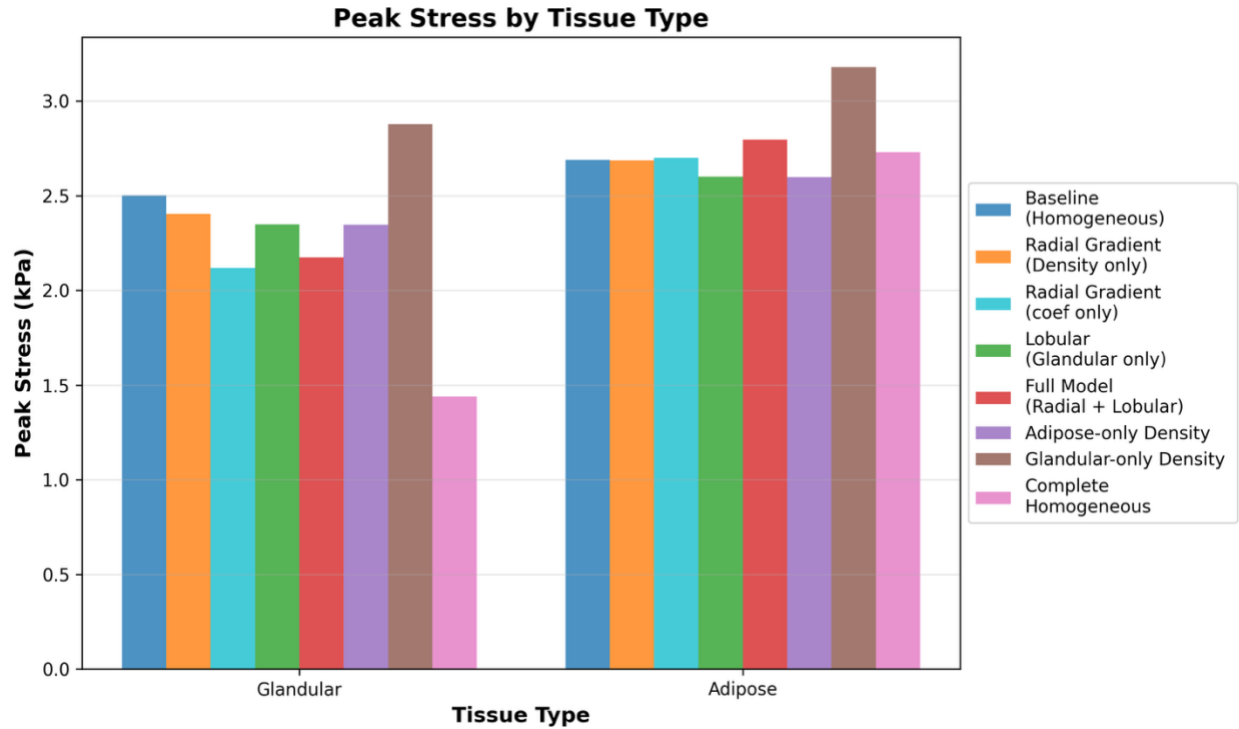


Figure 9: Peak von Mises stress by tissue type across eight model configurations. Glandular tissue shows greater stress variation between models (1.45-3.15 kPa) than adipose tissue (2.60-3.15 kPa). The Glandular-only density model exhibits the highest peak stress in glandular tissue.

The temporal evolution of stress showed tissue specific patterns throughout the simulation. The temporal evolution for the Full Model is shown in Figure 10a, with evolution patterns for all model configurations provided in Appendix B Figures 10b-10i. Glandular tissue maintained consistently higher mean stress than adipose tissue across all phases, with peak values of ~920 Pa versus ~675 Pa at jump initiation ($t = 1.025$ s). Both tissues followed similar temporal dynamics, with sharp increases at jump initiation, minimum values during upward acceleration (~1.1 s), and gradual recovery to steady state levels.

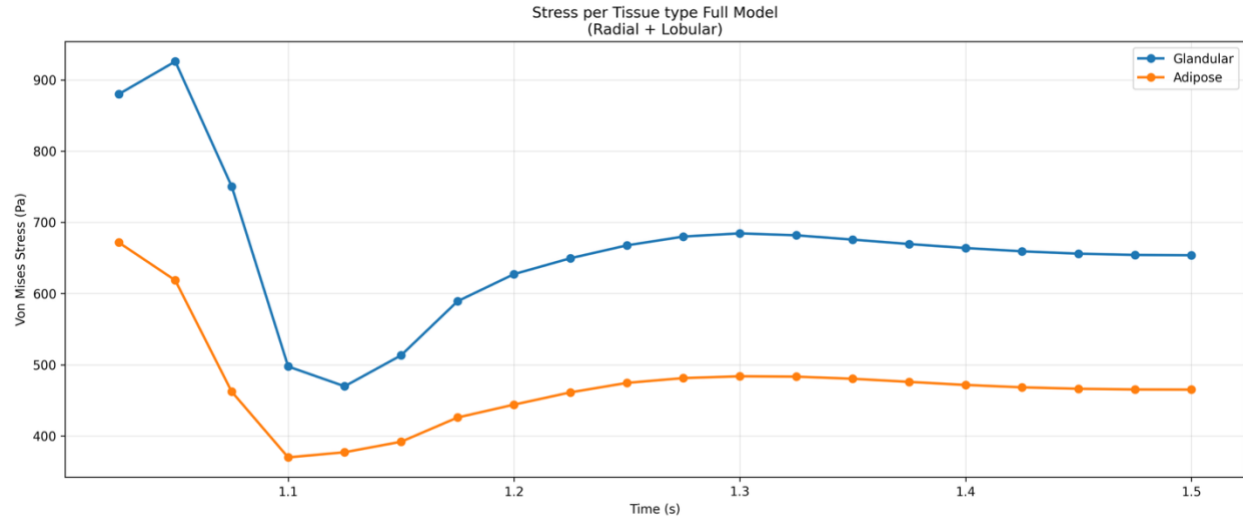


Figure 10a: Von Mises stress evolution for Full model during parabolic jump showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases, with peak values of ~920 Pa and ~675 Pa respectively at jump initiation ($t \approx 1.025$ s). Stress evolution for other model configurations shown in Appendix B Figures 10b-10h.

Discussion

This study investigated how spatial heterogeneity in tissue material properties affects breast biomechanical response during dynamic loading, with implications for optical imaging-based cancer detection systems. By implementing both radial density gradients and discrete glandular lobular structures within a finite element framework, this study assessed whether tissue heterogeneity produces detectable changes in surface displacement patterns and internal stress distributions.

Principal Findings

Heterogeneity Effects on Displacement

The introduction of spatial heterogeneity produced notable changes in surface displacement magnitude and spatial distribution compared to the homogeneous baseline model. Stiffness gradients (c_1 , c_2) reduced peak vertical displacement by 13% (14.0 mm to 12.2 mm), while density gradients had minimal effect (2% reduction, to 13.7 mm). This differential response indicates that elastic moduli, rather than tissue density, are the primary drivers of mechanical behaviour during gravitational loading and during physiologically relevant motions such as the parabolic jump scenario.

Models incorporating stiffness heterogeneity (Radial coef, Full Model) showed substantially smaller mean surface displacements (12.97-13.61 mm) compared to the Baseline (18.06 mm). The

spatial displacement patterns showed anterior concentration (35-45 mm) with radial decay toward the posterior chest wall (0-10 mm).

The position dependent response to material heterogeneity has important implications for displacement-based tumour detection systems. The nipple region showed an 89% increase in displacement in response to a 28% stiffness gradient, indicating strong mechanical amplification in areas with complex geometry. Since malignant tumours exhibit 200-500% stiffness increases relative to surrounding tissue, they should produce readily detectable displacement anomalies, particularly if located in mechanically sensitive regions.

The superior landmark's moderate consistency (5.95-6.76 mm range, within $\pm 5\%$ of baseline for most models) The relatively stable displacement at this location occurs in a region where the effects of gravitational loading and proximity to the chest wall boundary combine, creating a region with reduced sensitivity to internal property variations.

The inferior landmark's high variability (60% variation range, 3.60-5.77 mm) indicates mechanical sensitivity to depth-dependent property gradients. This position experiences cumulative effects from all tissue layers between the chest wall and anterior surface, with the radial stiffness gradient particularly affecting deformation (5.77 mm, +21% from baseline). This sensitivity makes the inferior region promising for detection property changes but requires careful baseline characterization to distinguish pathological from normal heterogeneity effects.

Stress Distributions

Peak von Mises stress analysis revealed differential tissue responses to heterogeneity. In glandular tissue, stiffness gradients reduced peak stress by 5% (from 2.22 kPa to 2.11 kPa) while density gradients increased it by 9% (to 2.41 kPa), demonstrating that the two heterogeneity components have opposing mechanical effects. In adipose tissue, both gradient types produced similar 17-18% stress reductions (from 3.23 kPa to 2.67-2.69 kPa), indicating that adipose mechanics are less sensitive to the specific form of heterogeneity.

These stress patterns, while not directly measurable by optical imaging, confirm that the heterogeneity implementation creates physiologically plausible mechanical loading states. The differential tissue responses confirm that the model captures the expected mechanical coupling between material properties and internal stress distributions.

Temporal Dynamics

All models exhibited three phase temporal behaviour: gravity equilibrium baseline, transient decrease during upward acceleration (with stress reductions of 45-50% from initial values of 695-735 Pa to approximately 380-470 Pa), and gradual recovery to steady state. Models with stiffness heterogeneity showed 4-7% lower initial stress levels and maintained smaller displacement magnitudes throughout, though temporal patterns remained consistent. This demonstrates that

heterogeneity affects displacement magnitude while preserving the fundamental dynamic response characteristics.

Comparison to Literature

Displacement Comparison

The Baseline model's peak vertical displacement of 14.0 mm during the parabolic jump falls well below the 42-110 mm range reported during running (McGhee & Steele, 2010). This difference reflects both the simplified loading protocol (a single 1 cm vertical jump versus continuous running gait) and the fixed chest wall boundary condition, which restricts motion compared to the compliant anatomical attachment.

Despite these limitations, relative displacement outcomes in finite element analyses of breast deformation have been quantified. Finite element studies of mammographic compression report mean nodal displacements of $\sim 11.6 \pm 10.6$ mm (mean Euclidean 3D displacement), demonstrating that whole-breast displacement fields can be meaningfully compared across simulation conditions (Martínez-Martínez et al., 2017).

In this study, the introduction of stiffness heterogeneity resulted in a 25% reduction in mean surface displacement (3.25 mm decrease from the 16.22 mm baseline), consistent with prior findings that local increases in tissue stiffness constrain surrounding tissue motion. However, direct quantitative comparison remains limited, as most prior work focuses on planar mammographic compression with large uniaxial strains, whereas the present model considers gravitational and dynamic loading.

Mechanical Behaviour Comparison

The tissue specific von Mises stress values observed in this study (adipose peak: 1.7–3.2 kPa; glandular peak: 1.4–2.9 kPa) fall within the range of stresses reported in recent finite element studies of breast biomechanics. Chen et al. (2024) reported peak equivalent von Mises stresses of approximately 2.9–3.4 kPa in glandular tissue within a multi component finite element breast model subjected to dynamic loading. In contrast much higher equivalent von Mises stresses, ranging from 8.0 to 15.9 kPa, have been observed in adipose tissue near glandular and Cooper's ligaments, highlighting the potential influence of local anatomical features on stress concentrations. Similarly, Chen et al. (2025) reported maximum von Mises stresses of 3.3–3.8 kPa in glandular tissue and 2.1–4.5 kPa in adipose tissue using a multi-component model under dynamic motion. This studies model predicts slightly lower peaks but remains within these ranges, supporting the physiological plausibility of the stresses despite differences in geometric complexity and loading conditions.

Implications for Cancer Detection

Surface Displacement Effects

The finding that spatial property variations create detectable surface displacement differences (up to 89% variability at the nipple, 21% at the inferior landmark) suggests that heterogeneity may enhance rather than obscure tumour detection in optical imaging systems. A tumour, represented as a localized stiffness increase, would create displacement anomalies superimposed on the background heterogeneity pattern.

Since the EWS optical imaging system captures surface motion from multiple angles simultaneously, the position dependent sensitivity observed here is not a limitation but rather an opportunity. The system can integrate displacement information from all visible regions, potentially triangulating the location and properties of internal stiffness anomalies from the pattern of surface displacement variations observed across different viewing angles. If the imaging system can establish a baseline heterogeneity signature for an individual patient through repeated measurements, deviations from this baseline could indicate new pathological changes.

Stiffness Contrast Requirements

The 13% displacement reduction observed with stiffness gradients (c_1 variation of 30 Pa over 109 Pa baseline, representing 28% stiffness increase) provides insight into detection sensitivity requirements. The 89% nipple displacement increase from a 28% stiffness gradient demonstrates notable mechanical amplification effects. Clinical studies report that malignant tumours exhibit 3-6× higher stiffness than surrounding tissue (Kim Joo Kim et al., 2025), representing 200-500% stiffness contrasts. Given that the relatively modest 28% stiffness gradient produced a 13% displacement change, tumours with their much larger stiffness contrasts should create readily detectable displacement anomalies. This suggests that tumours should be readily detectable through surface displacement analysis.

Measurement Strategy

The differential response of surface displacement (13% reduction with stiffness gradients) versus internal stress (5-17% changes depending on tissue type) demonstrates that surface displacement measurements capture mechanical property changes, although the relationship is complex. For the optical imaging system, this implies that surface displacement is indeed sensitive to internal tissue configuration and should respond to the presence of a tumour. However, the mapping between displacement and stiffness is nonlinear and position dependent, meaning that calibration is essential for accurate interpretation. Training AI with internal tissue data could help it detect tumours by mapping observed surface motion back to underlying material properties.

The position specific variability patterns observed provide practical guidance for optical imaging system design. The nipple and inferior regions' high sensitivity (89% and 21% displacement

variability) make them promise for detecting property changes. Monitoring both mechanically sensitive and stable regions could optimize detection accuracy by providing internal calibration for displacement measurements.

Limitations

Geometric and Material Simplifications

The idealized hemispherical, axisymmetric geometry does not capture individual anatomical variations, the inframammary fold where breast meets chest wall, or surrounding structures as the sternum and ribcage. The isotropic Mooney-Rivlin material model represents tissue as having uniform properties in all directions, while real breast tissue exhibits directional dependencies in mechanical response. These simplifications were necessary for computational tractability and to isolate the specific effects of heterogeneity, but they limit direct applicability to patient specific scenarios.

The heterogeneity patterns (smooth radial gradients, Gaussian lobules) represent idealized mathematical distributions chosen for parametric control rather than actual tissue microstructure. Real tissue exhibits irregular branching networks and complex tissue boundaries that are difficult to capture with simple analytical functions. Future work incorporating medical imaging data could provide more anatomically realistic property distributions.

Boundary Conditions and Loading Protocol

The fixed chest wall boundary does not account for underlying tissue compliance and fascial sliding, which may lead to overestimating stress concentrations at this interface and underestimating overall breast displacement. The prescribed parabolic displacement represents simplified vertical-only motion, omitting the multi-directional kinematics, skin sliding, and muscle activation present during real activities.

The loading protocol was designed to produce measurable deformation while ensuring numerical stability. The 1 cm maximum displacement represents a mild loading scenario, substantially lower than the displacements during vigorous activities such as running or jumping. This explains the lower absolute displacement values compared to clinical studies but does not invalidate the relative comparison between heterogeneous model variants, which is the primary focus of this study.

Parameter Uncertainty and Validation

Material parameters (c_1 , c_2 , ρ , κ) were derived from published literature values representing population averages rather than measured from specific individuals. The heterogeneity amplitude parameters ($\alpha_{c_1} = 30$ Pa, lobule amplitudes of 70 Pa) were selected to produce property variations

within reported ranges (20-40% coefficient of variation in elastography studies; McKnight et al., 2002; Patel et al., 2020).

Direct validation against clinical data was not performed, as maps of internal tissue properties and mechanical response measurements from the same individuals were unavailable. Without such validation, model predictions remain theoretical. Establishing confidence in predictive capability would require experimental protocols measuring both internal tissue properties (via elastography) and surface displacement (via optical tracking) during controlled loading.

Future Directions

The next step for advancing model realism is implementing free form geometries, combined with spatially resolved material property maps derived from MRI or CT imaging. Modern MRI sequences can provide both anatomical details, such as adipose and glandular tissue boundaries, and mechanical property distributions through MR elastography. Integrating these data would allow the model to capture patient specific anatomy, including breast shape, size, chest wall curvature, nipple position, and internal tissue distributions. It would also allow the use of measured stiffness maps instead of literature-based averages, reducing uncertainty. Furthermore, patient specific validation would become possible by comparing model predictions against measured surface displacement from the same individual.

Incorporating Cooper's ligament structures represents an important anatomical feature currently missing from the model. These fibrous bands provide primary structural support and create mechanical coupling between skin and deeper tissue, fundamentally influencing breast deformation patterns. Implementation could proceed through adding a network of cable like elements representing major ligaments, positioned based on anatomical literature or extracted from MRI images. Cooper's ligaments would enable investigation of how fibrous structure interacts with the volumetric tissue heterogeneity already implemented, providing a more complete representation of breast mechanical architecture. To further improve boundary realism, future work could also consider replacing the current rigid chest wall with a deformable pectoralis major muscle layer. Including this muscle, modelled with transversely isotropic properties to capture fiber directionality, would provide more physiologically accurate constraints and could alter stress and displacement patterns, as demonstrated by Chen et al. (2025). Optionally, the ribcage could remain as a rigid posterior boundary to preserve structural support.

The heterogeneity framework developed in this study provides a foundation for investigating tumour detection in more realistic tissue environments. Future work should incorporate localized stiffness inclusions representing tumours of varying sizes and positions within the heterogeneous tissue model, building on recent advances in tumour modelling complexity (Diepstraten, 2026). By simulating tumours embedded within the heterogeneous architecture implemented here, future studies can assess how natural tissue variation influences tumour detectability. These results

should be compared to those obtained in homogeneous baseline model to evaluate the impact of heterogeneity on surface displacement analysis.

Together, these simulation-based approaches would transform the finite element model from a proof-of-concept tool into a practical training data generator for machine learning algorithms. Clinical datasets are inherently limited in size and cannot systematically cover the full spectrum of anatomical variations and pathological conditions. Simulations with precisely known ground truth tissue properties would enable systematic validation of inverse models and comprehensive coverage of anatomical variations, including rare pathological cases, before clinical trials.

Conclusion

This study addresses the central research question of how incorporating tissue heterogeneity and glandular lobular structure affects the biomechanical response of the breast model during simulated motion and how these changes manifest in observable surface displacement patterns. Current finite element breast models often assume homogeneous tissue properties, failing to capture the anatomical complexity where adipose and glandular tissues differ substantially in density and stiffness with coefficients of variation of 20-60% within tissue types (McKnight et al., 2002; Goodbrake et al., 2022; Patel et al., 2020). This study directly addressed this limitation by implementing spatially varying mechanical properties representing natural tissue variation.

The results demonstrate that stiffness gradients, rather than density variations, primarily drive surface displacement patterns (13% reduction versus 2% reduction), providing crucial insight for imaging system development: observable surface deformations are indeed sensitive to internal tissue property distributions. The position dependent sensitivity, particularly the 89% displacement increase at the nipple with stiffness gradients, confirms that different breast regions respond differently to internal heterogeneity.

Most significantly for cancer detection applications, the 13% displacement change produced by a relatively modest 28% stiffness gradient indicates that malignant tumours with their substantially higher stiffness contrasts (200-500% above surrounding tissue) should create displacement anomalies large enough to be detected through surface displacement analysis. This finding directly supports the feasibility of the optical imaging approach: biomechanical differences between malignant tumours and healthy tissue could be captured through synchronized multi camera imaging of surface motion patterns.

The heterogeneous model framework developed in this study demonstrates that tissue property variations produce measurable surface displacement differences. This finding confirms that heterogeneity must be incorporated in models supporting cancer detection systems. The current implementation, based on idealized geometry and literature-derived parameters, serves as a proof of concept rather than a practical AI training tool. However, it establishes a computational

foundation for future patient specific implementations. These will enable systematic investigation of how tissue property variations affect observable displacement patterns.

Incorporating tissue heterogeneity fundamentally alters the breast model's biomechanical response in ways that are both measurable at the surface and consistent with physiological tissue behaviour. The displacement patterns and stress distributions observed validate that the heterogeneous implementation creates mechanically realistic loading states while maintaining reasonable computation time. This enhanced model serves as a validated foundation for generating synthetic training data and supports the continued development of radiation free breast cancer detection technology. Future work incorporating patient specific geometries from medical imaging, Cooper's ligament structures, and active muscle mechanics will further improve the model's clinical applicability, ultimately contributing to more effective early breast cancer detection for women across all age groups and tissue densities.

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Appendix A

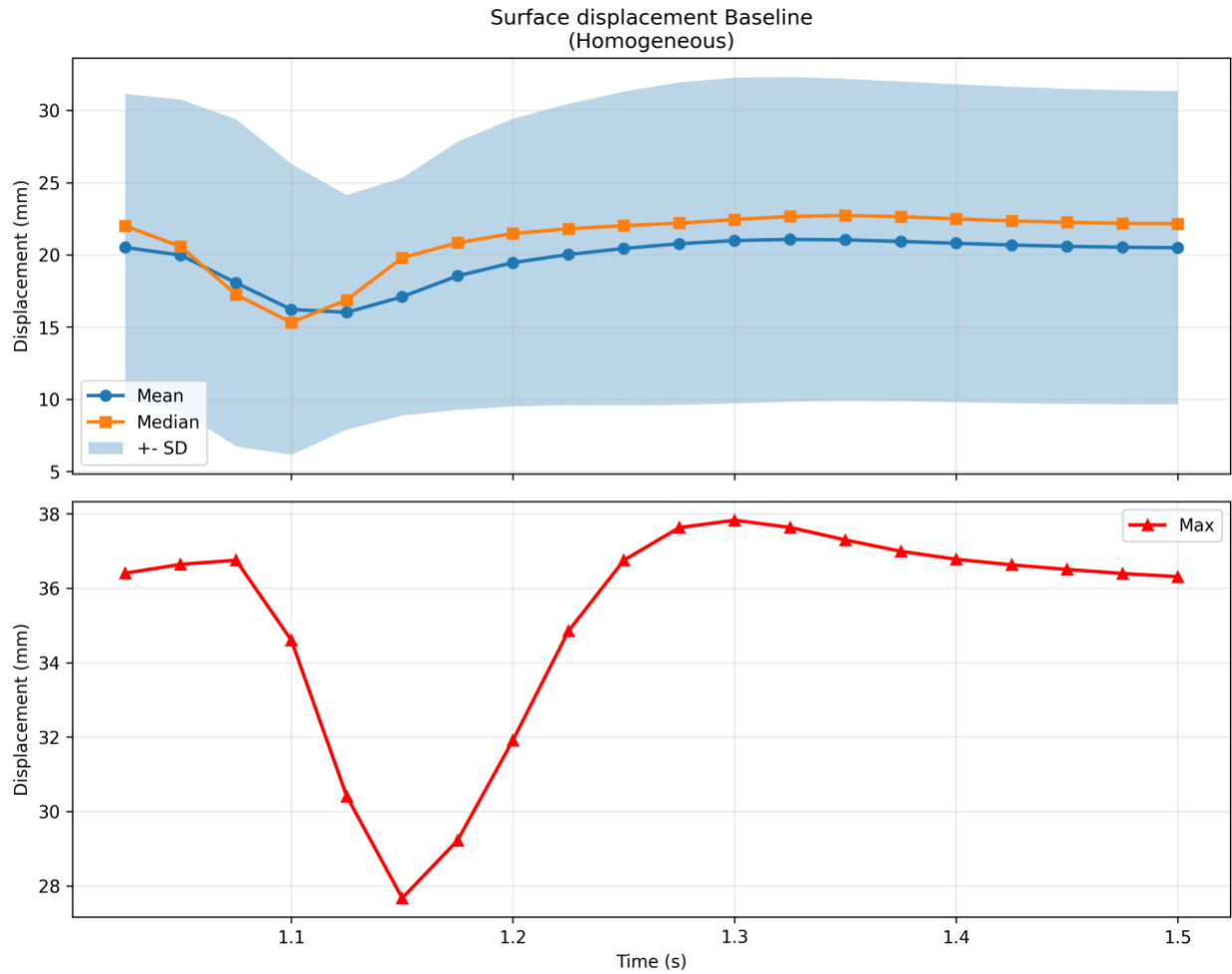


Figure 6b: Surface displacement evolution for Baseline homogeneous. Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

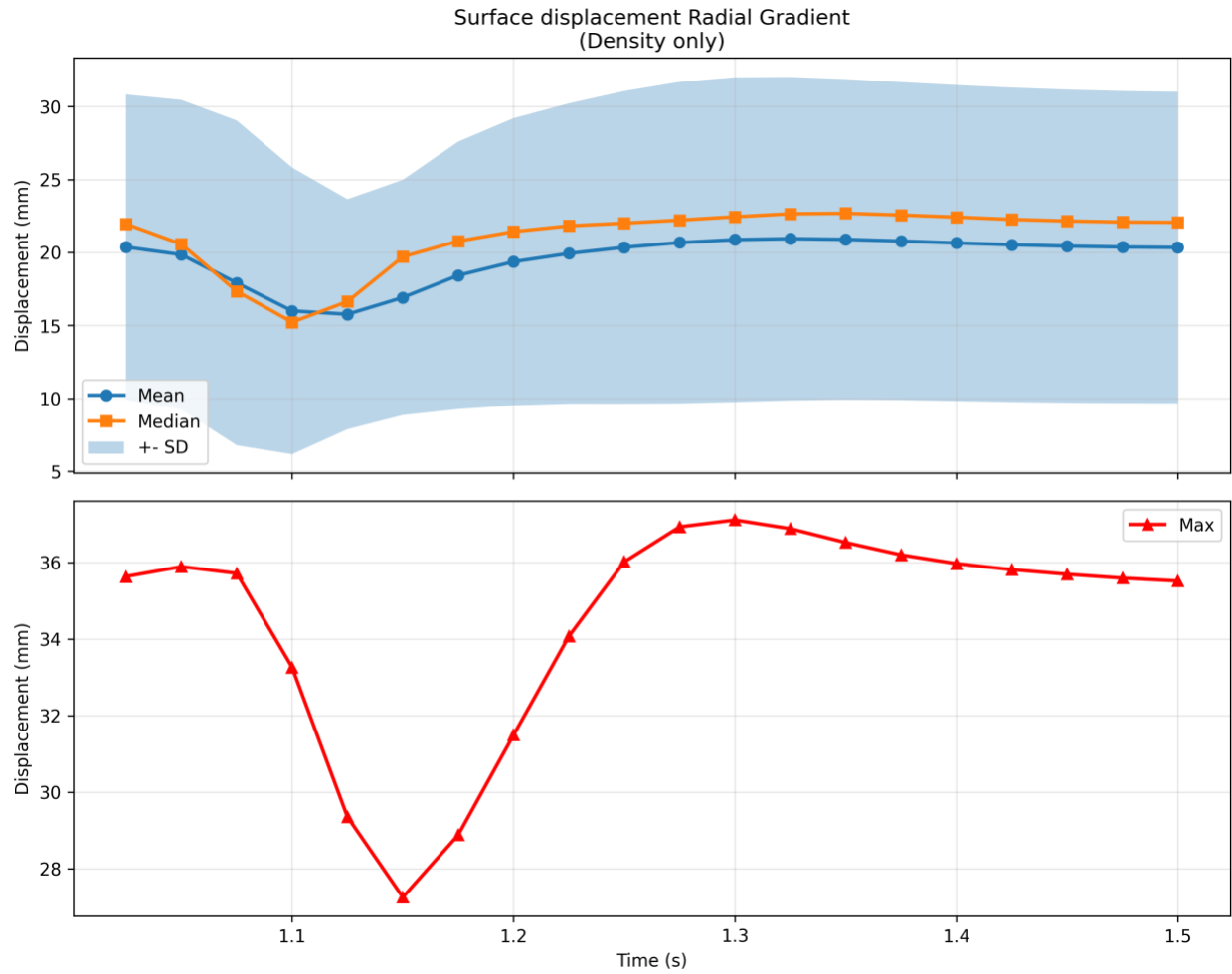


Figure 6c: Surface displacement evolution for Radial Gradient (Density only). Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

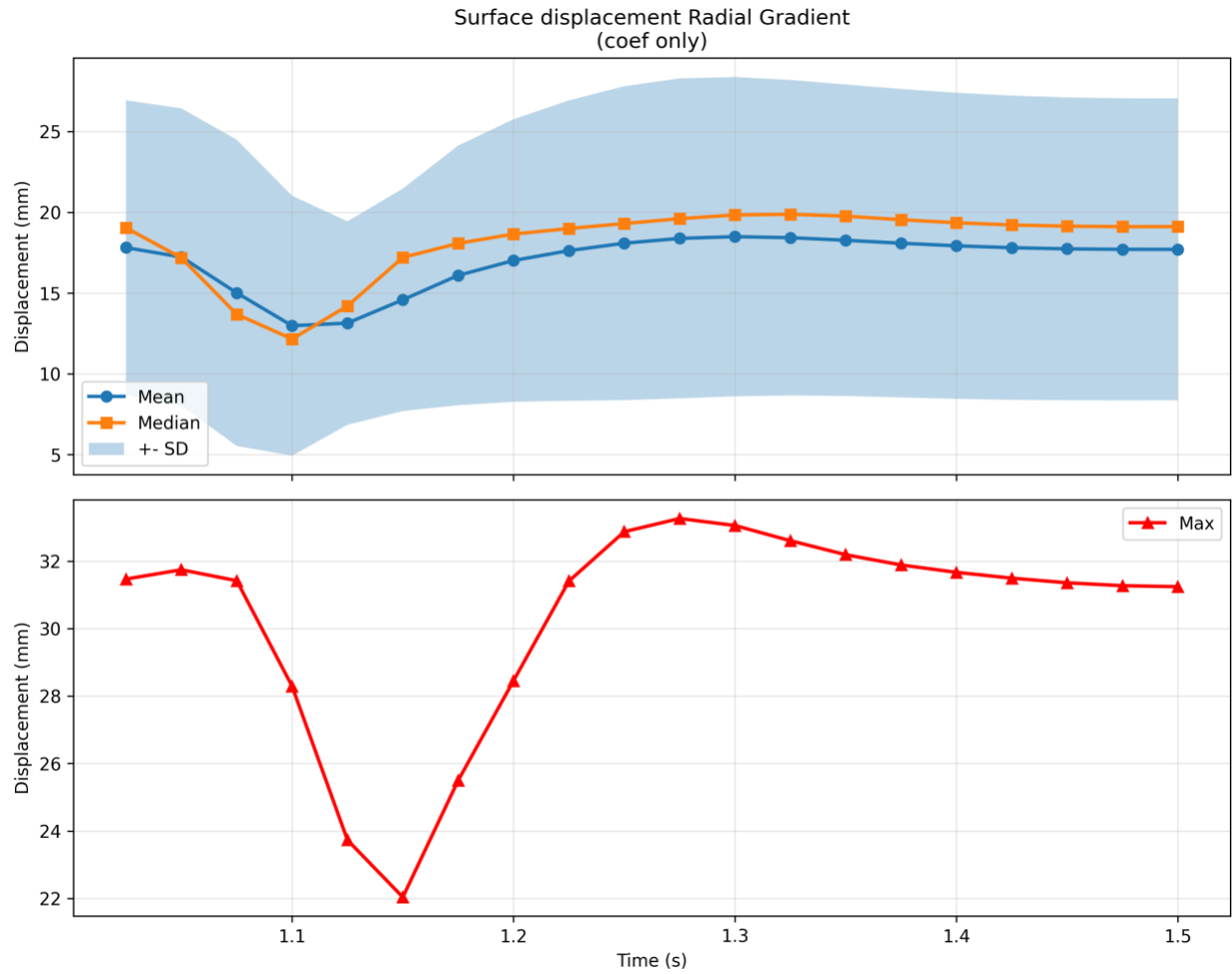


Figure 6d: Surface displacement evolution for Radial Gradient (coef only). Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

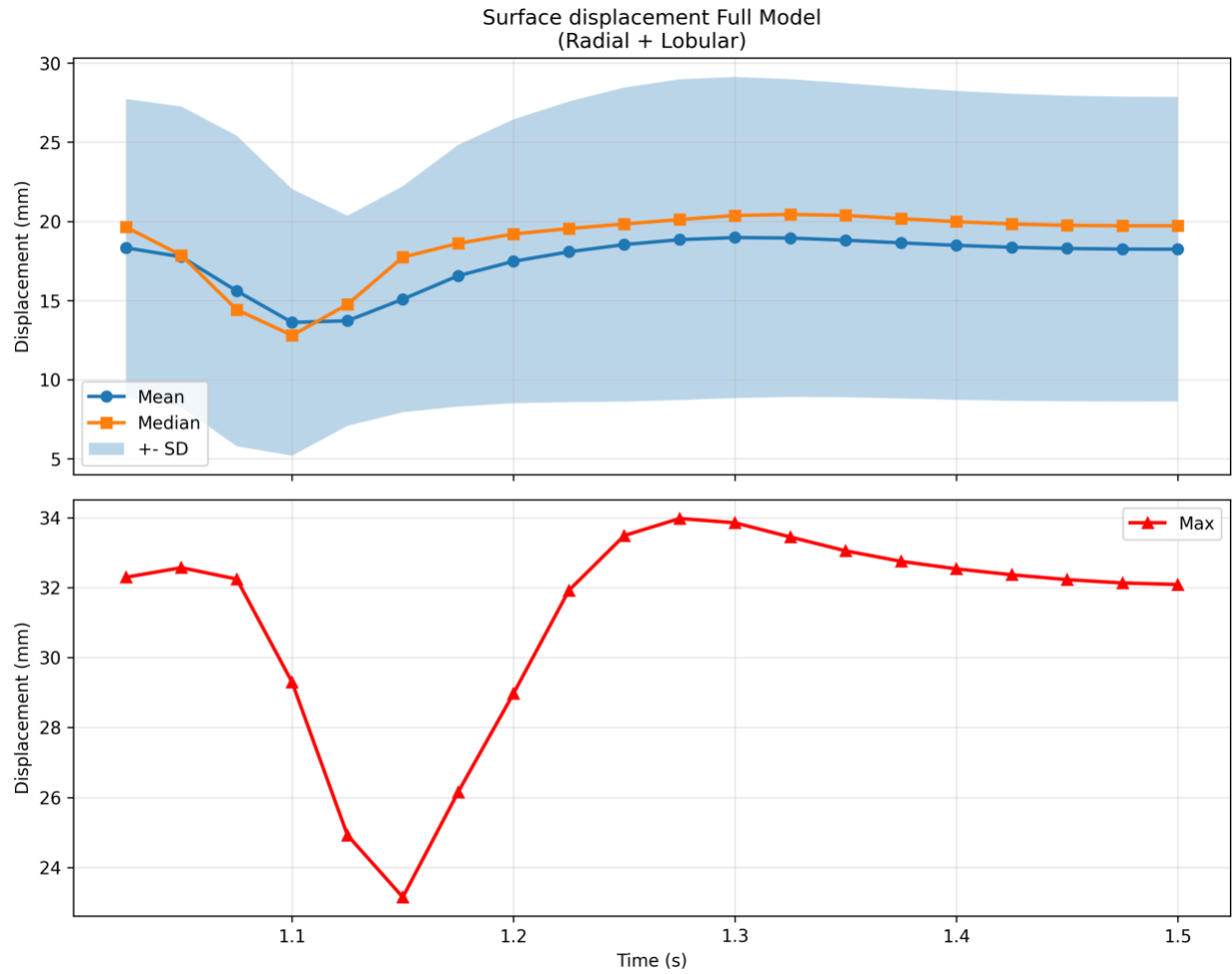


Figure 6e: Surface displacement evolution for Full Model (Radial + Lobular). Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

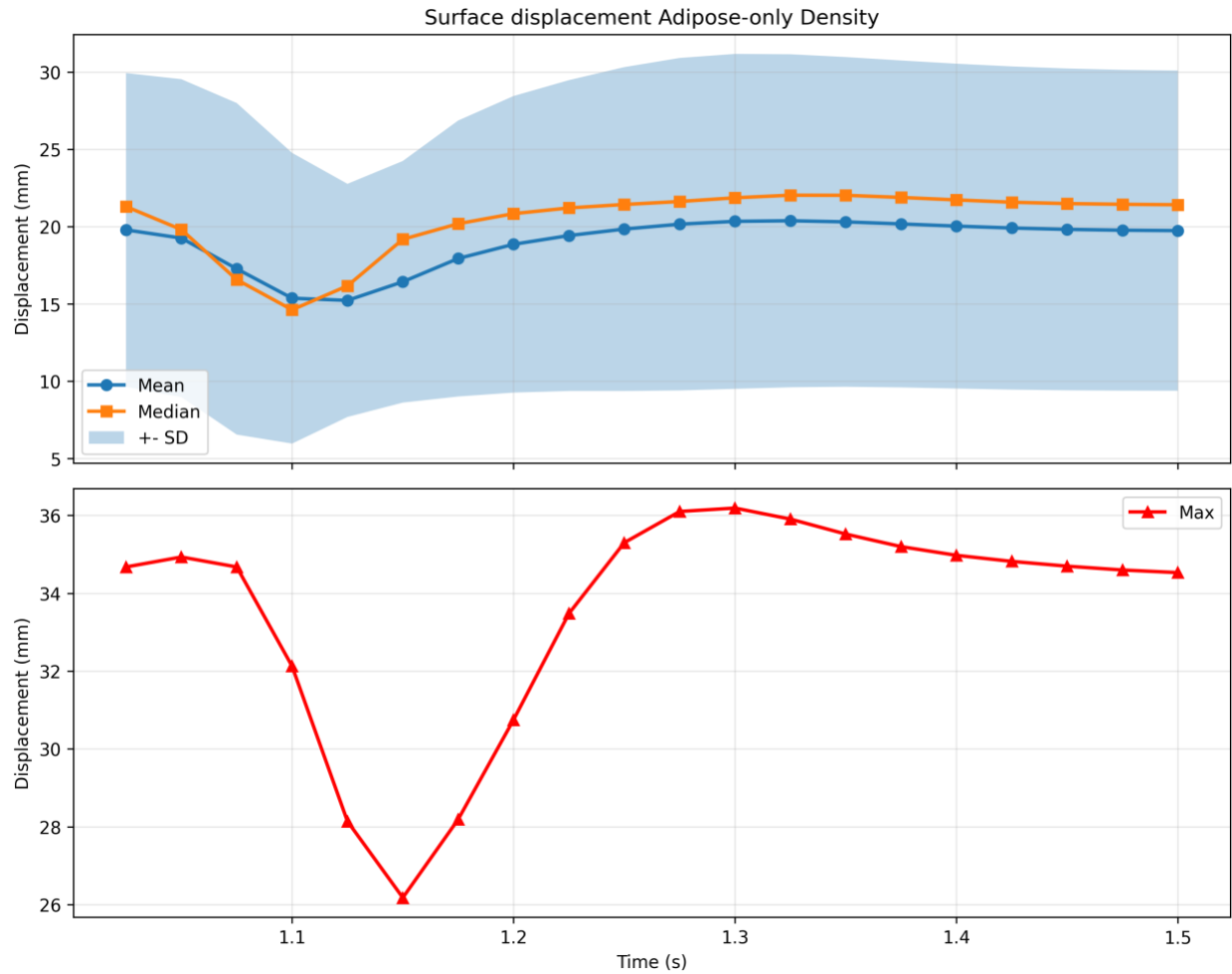


Figure 6f: Surface displacement evolution for Adipose-only Density. Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

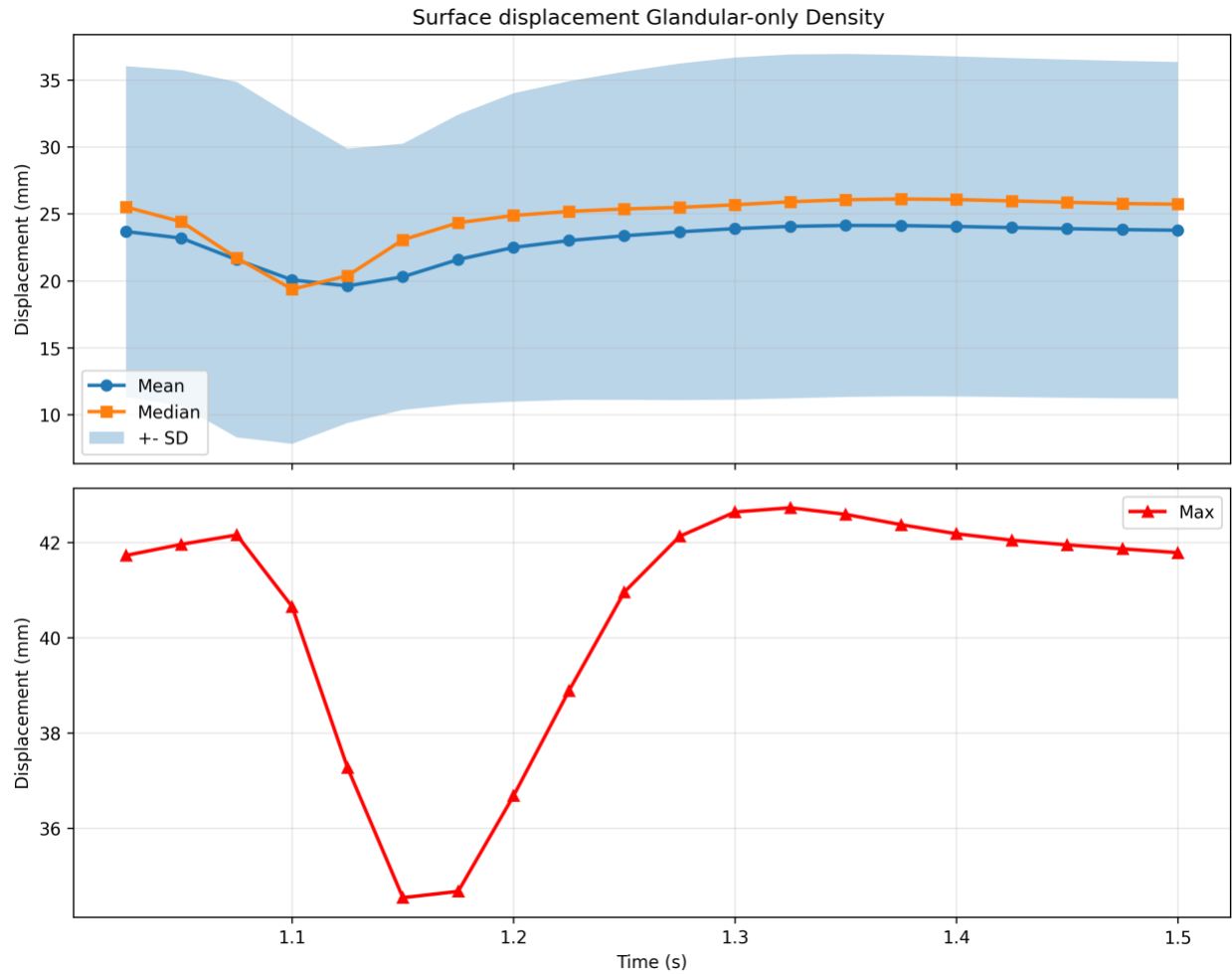


Figure 6g: Surface displacement evolution for Glandular-only Density. Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

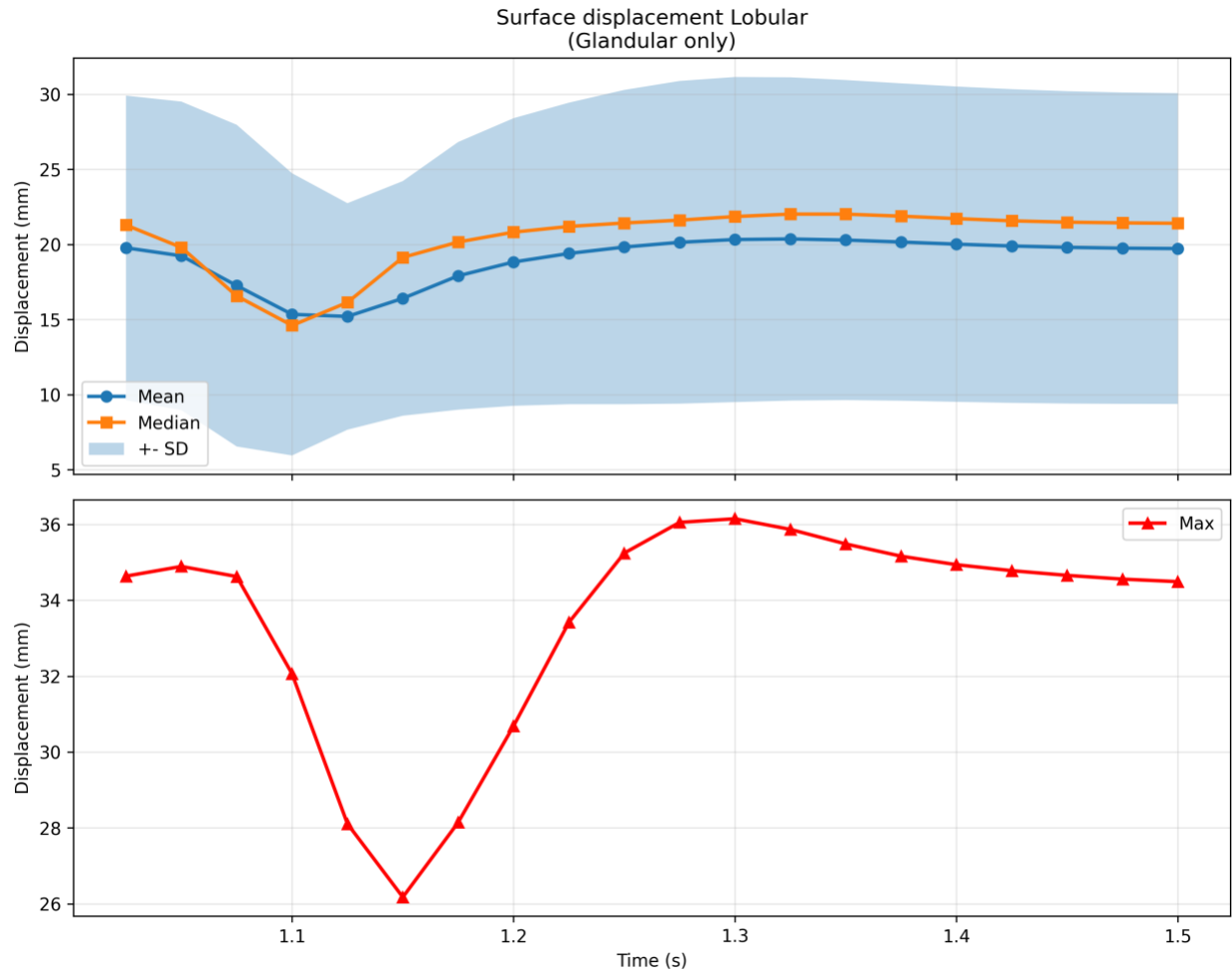


Figure 6h: Surface displacement evolution for Lobular (Glandular only). Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

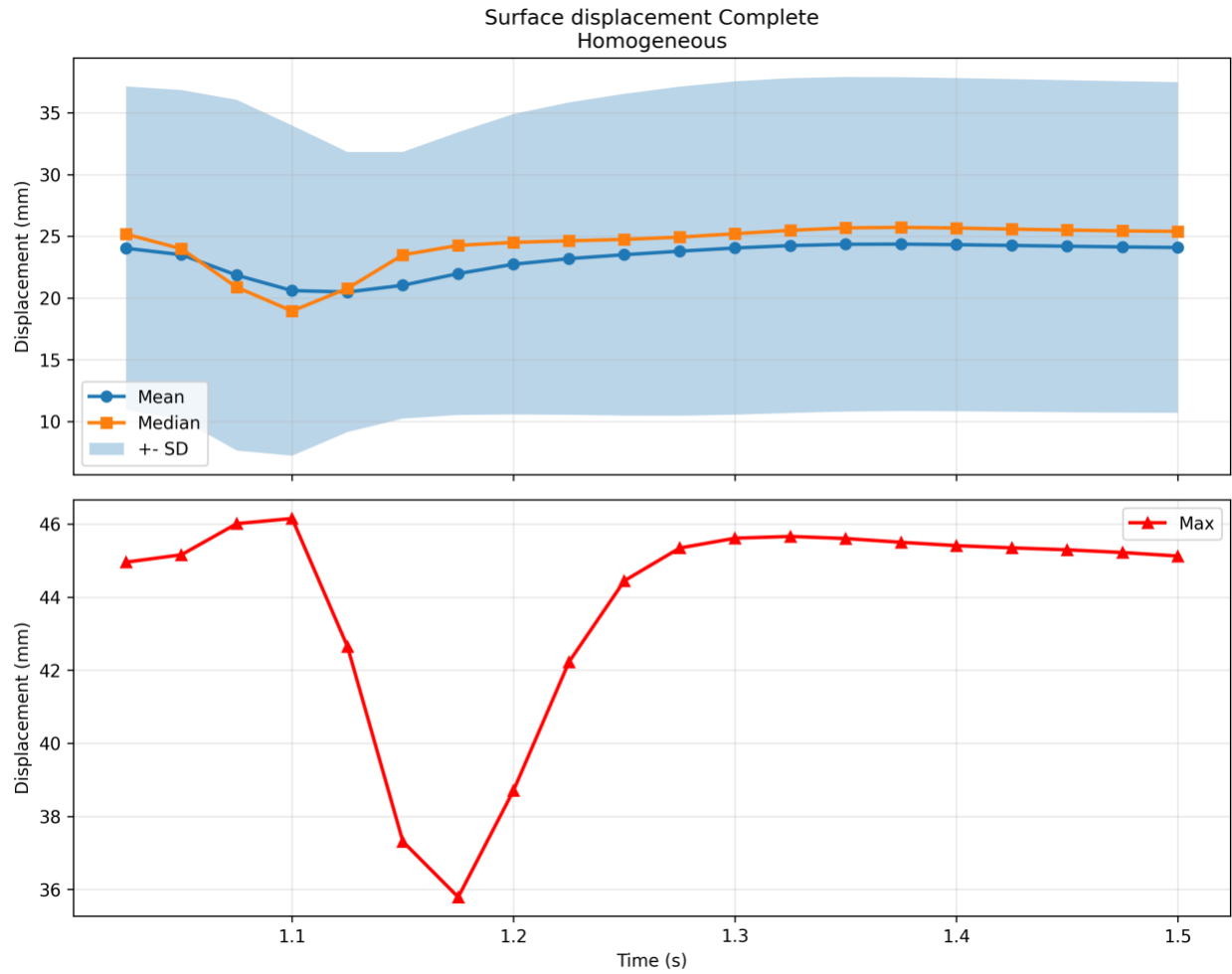


Figure 6i: Surface displacement evolution for Complete Homogeneous. Top panel: mean (blue), median (orange), and SD range (shaded) showing absolute displacement from undeformed state. Bottom panel: maximum displacement across all surface nodes. Models exhibit three phase behaviour: (1) gravity equilibrium baseline ($t \approx 1.025$ s), (2) transient decrease during upward acceleration ($t \approx 1.1$ s), and (3) recovery to new equilibrium ($t > 1.2$ s).

Appendix B

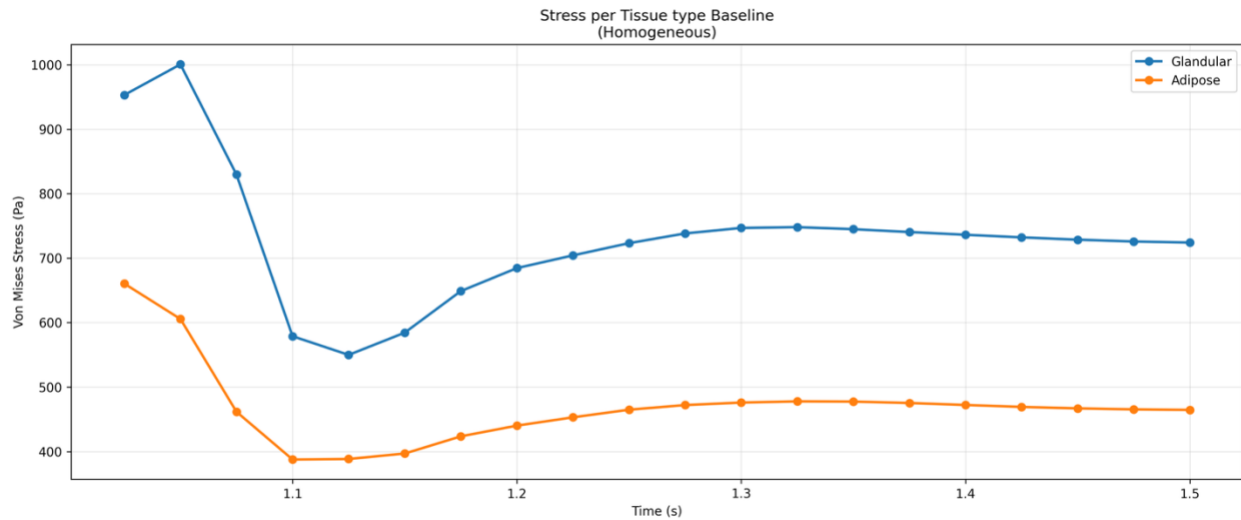


Figure 10b: Von Mises stress evolution for Baseline (Homogeneous). During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~1000 Pa and adipose stress around ~675 Pa.

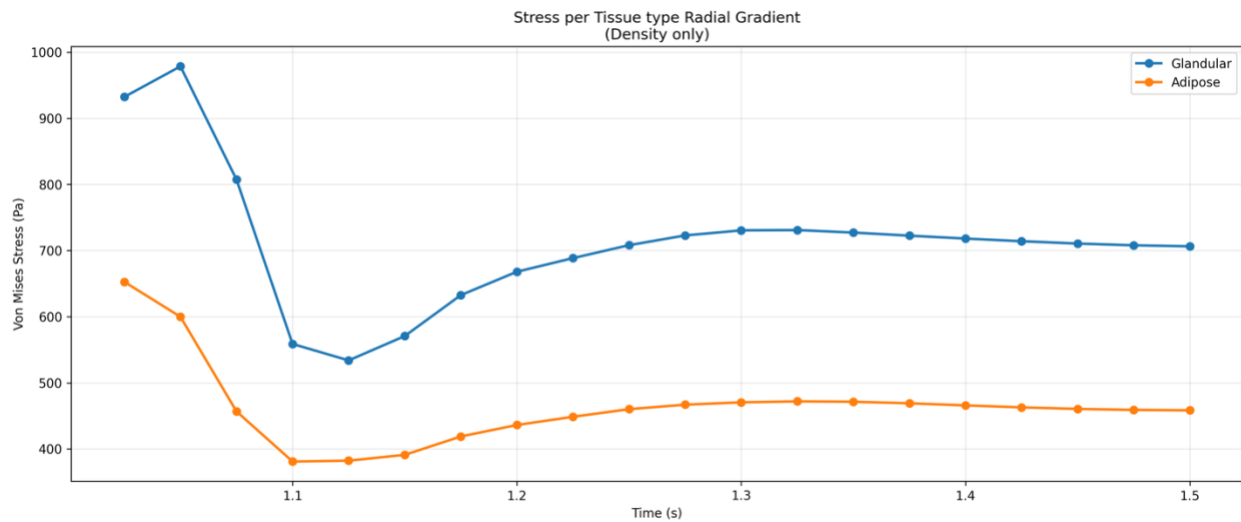


Figure 10c: Von Mises stress evolution for Radial Gradient (Density only). During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~980 Pa and adipose stress around ~655 Pa.

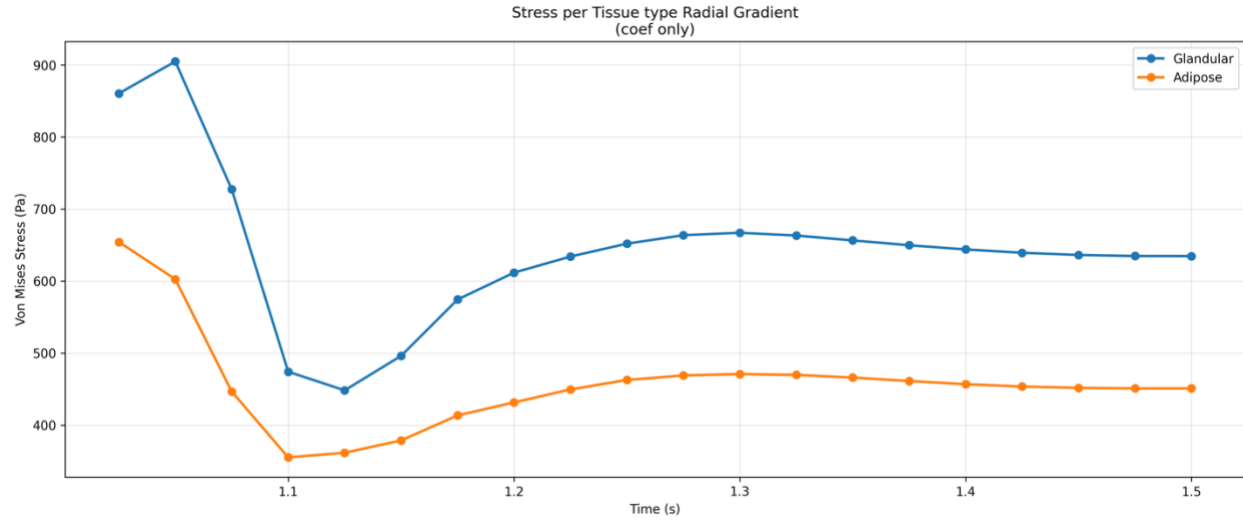


Figure 10d: Von Mises stress evolution for Radial Gradient (coef only). During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~905 Pa and adipose stress around ~650 Pa.

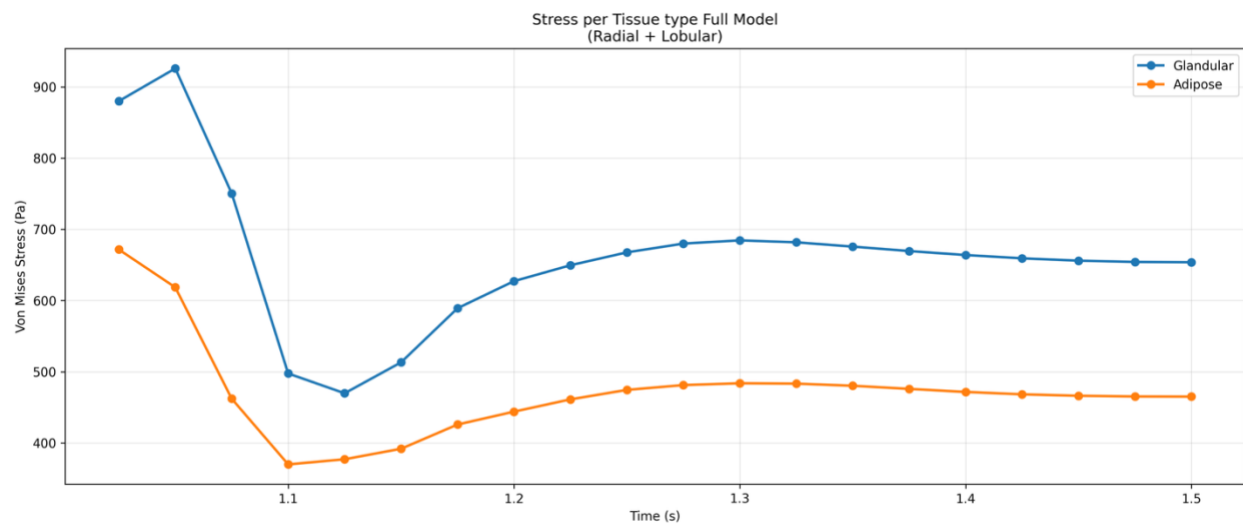


Figure 10e: Von Mises stress evolution for Full Model (Radial + Lobular). During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~920 Pa and adipose stress around ~680 Pa.

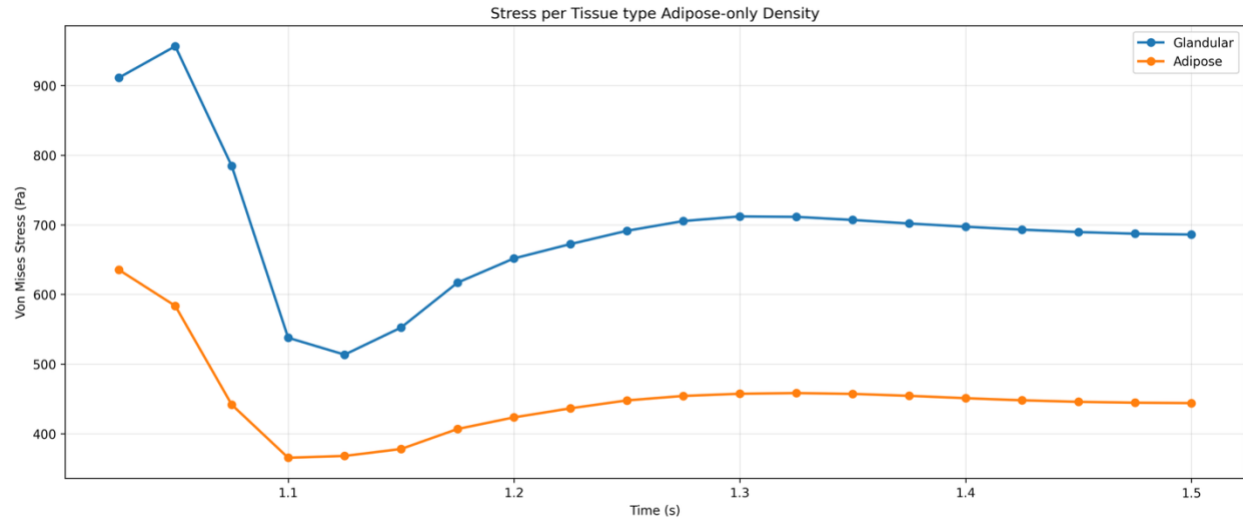


Figure 10f: Von Mises stress evolution for Adipose-only Density. During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~975 Pa and adipose stress around ~630 Pa.

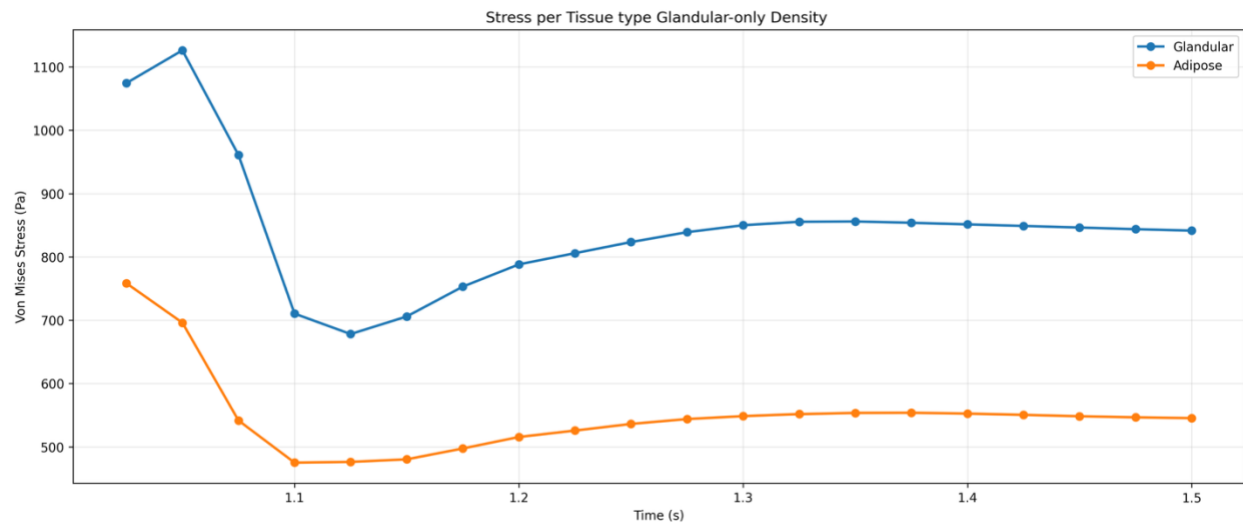


Figure 10g: Von Mises stress evolution for Glandular-only Density. During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~1120 Pa and adipose stress around ~760 Pa.

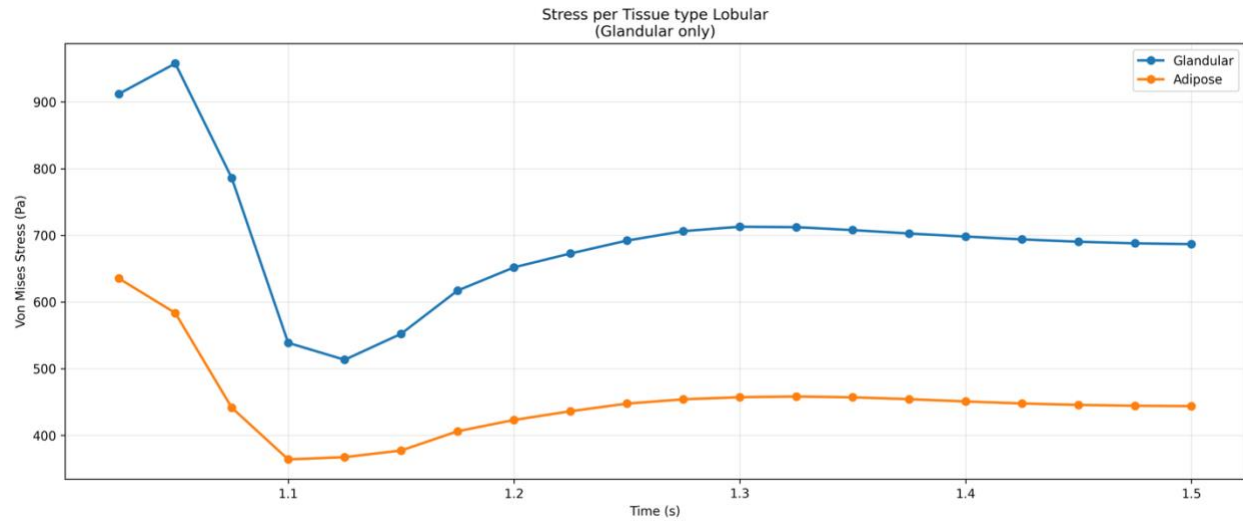


Figure 10h: Von Mises stress evolution for Lobular (Glandular only). During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~980 Pa and adipose stress around ~640 Pa.

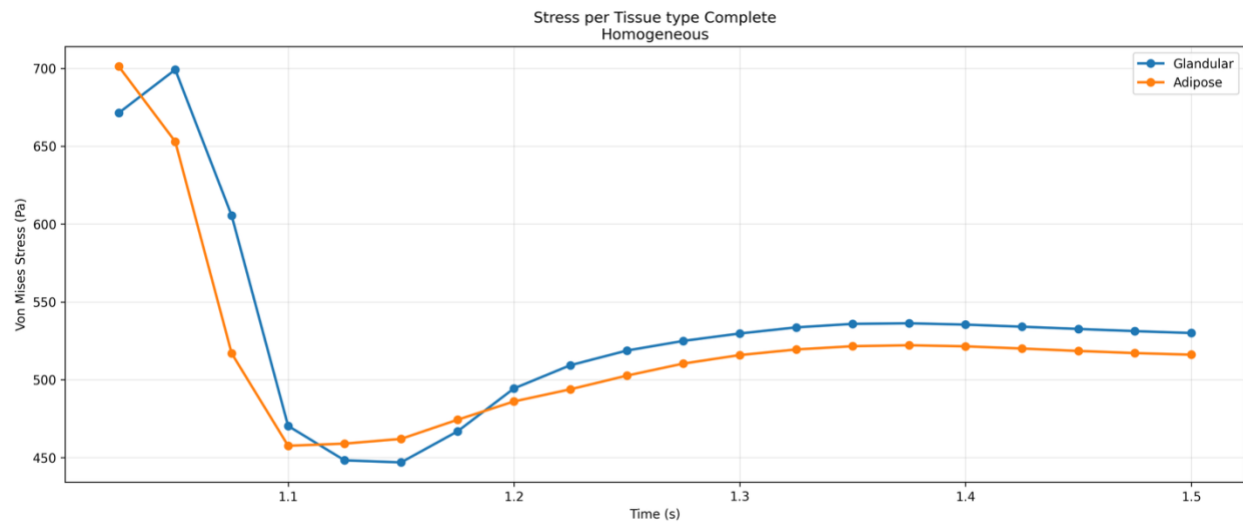


Figure 10i: Von Mises stress evolution for Complete Homogeneous. During parabolic jump, showing tissue specific responses. Glandular tissue (blue) maintains consistently higher stress than adipose tissue (orange) throughout all phases. Peak values occur at jump initiation ($t \approx 1.025$ s), with glandular stress around ~700 Pa and adipose stress around ~705 Pa.